A REAL K3 AUTOMORPHISM WITH MOST OF ITS ENTROPY IN THE REAL PART

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Abstract. This article describes the first example of a real K3 surface X admitting a real automorphism f satisfying $h_{top}(f, X(\mathbb{C})) < 2h_{top}(f, X(\mathbb{R}))$. The example presented is a $(2,2,2)$-surface in $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ defined by the vanishing set of $(1+x^2)(1+y^2)(1+z^2)+10xyz-2$, first described by McMullen. Along the way, we develop an ad hoc shadowing lemma for C^2 (real) surface diffeomorphisms, and apply it to estimate the location of a periodic point in $X(\mathbb{R})$. This result uses the GNU MPFR arbitrary-precision arithmetic library in C and the Flipper computer program.

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1. Introduction

Let X be a K3 surface equipped with a real structure, and let $f : X \to X$ be a *real automorphism* of X; that is, a holomorphic diffeomorphism which commutes with the given antiholomorphic involution σ of X. The fixed point set of σ, denoted $X(\mathbb{R})$, is either empty or a (possibly disconnected) compact, real-analytic, totally real submanifold with real dimension two. Trivially, $h_{top}(f, X(\mathbb{R})) \leq h_{top}(f, X(\mathbb{C}))$. However, in his ICM survey ([4]), Cantat noted that there were no known examples for which $h_{top}(f, X(\mathbb{C})) < 2h_{top}(f, X(\mathbb{R}))$. The purpose of this paper is to provide such an example.

Before beginning, we recall a few related facts. Cantat proved in [2] that if Y is a connected compact complex surface equipped with an automorphism f of positive topological entropy, then Y is Kähler and one of the following holds:

(1) either Y is a non-minimal rational surface;
(2) or Y is a K3 surface, a complex torus, or an Enriques surface;
(3) or there exists a surface $\tilde{Y}$ from case (2) and a holomorphic birational map $\pi : Y \to \tilde{Y}$ such that $\tilde{f} \coloneqq \pi \circ f \circ \pi^{-1}$ is an automorphism of $\tilde{Y}$.

Given a surface Y with a real structure and a real automorphism f with positive entropy, we let $\rho(f) \coloneqq \frac{h_{top}(f, Y(\mathbb{R}))}{h_{top}(f, Y(\mathbb{C}))}$ denote the ratio of the real and complex entropies.

In case (3), the surface Y is obtained from $\tilde{Y}$ by a finite sequence of $\tilde{f}$-equivariant blow-ups along periodic orbits of $\tilde{f}$. Since the entropy of an automorphism of a Kähler manifold is determined by the spectral radius of its action on cohomology (see the Gromov–Yomdin theorem [9], [10], [21]), $h_{top}(f, Y(\mathbb{C})) = h_{top}(\tilde{f}, \tilde{Y}(\mathbb{C}))$. Moreover, even if some of the blow-ups are along an orbit in the real part, one can show $h_{top}(f, Y(\mathbb{R})) = h_{top}(\tilde{f}, \tilde{Y}(\mathbb{R}))$. Therefore, it's enough to focus on cases (1) and (2).

In the case that Y is a rational surface, Diller-Kim produced a family of examples for which $\rho(f) = 1$ ([6], Theorem A). They also provided examples for which $\frac{1}{2} < \rho(f) < 1$ ([6], Theorem B). To my knowledge, in the rational case, no examples are known for which $\rho(f) = 0$.

For K3 surfaces, it remains open whether $\rho(f) = 1$ or $\rho(f) = 0$ can occur. On one hand, Moncet proved that for any $\varepsilon > 0$, there exists a K3 surface admitting a real automorphism f with $\rho(f) < \varepsilon$

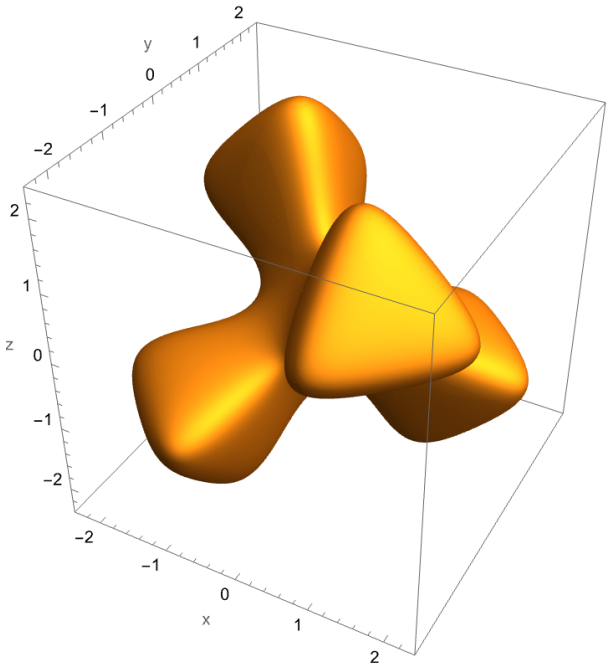


FIGURE 1. $X_{10}(\mathbb{R})$, plotted in Mathematica

([16], Theorem 5.7). He also showed that if Y is a complex torus, then any real automorphism f of Y with positive entropy satisfies $\rho(f) = \frac{1}{2}$ ([16], Proposition 4.4). The same holds when (Y, f) is a so-called Kummer example in the sense of [5]. For instance, a K3 Kummer example (Y, f) is obtained by taking Y to be the minimal regular model of the quotient of the standard complex torus $\mathbb{C}^2/\Lambda$ by $(x, y) \mapsto (-x, -y)$, and taking f to be induced by a real hyperbolic automorphism of the torus.

Generally, a few strategies for bounding $\rho(f)$ from below have emerged. While complex entropy can usually be computed explicitly using the Gromov–Yomdin theorem, the challenge becomes providing a lower bound on the real entropy. In [6], Diller-Kim bounded the real entropy by understanding how f acts on the homology of the real part. In forthcoming work, Kim provides sharper lower bounds on $\rho(f)$ for some automorphisms by analyzing homotopy growth rates. Alternatively, Moncet proved that $\rho(f)$ is bounded below by an intrinsic property of the surface Y called its "concordance" ([16]). However, concordance is generally tricky to calculate. In the present work, we bound real entropy by computing the stretch factor of the mapping class of f, acting on the real surface with punctures along a real periodic orbit.

1.1. **Setup and Main Theorem.** Given nonzero $A \in \mathbb{R}$, consider

$$q_A(x, y, z) \coloneqq (1 + x^2)(1 + y^2)(1 + z^2) + Axyz - 2.$$

Taking $([x_0 : x], [y_0 : y], [z_0 : z])$ to be homogeneous coordinates for $(\mathbb{P}^1)^3$, one has

$$q_A(x, y, z) = \tilde{q}_A([1 : x], [1 : y], [1 : z])$$

where

$$\tilde{q}_A([x_0 : x], [y_0 : y], [z_0 : z]) \coloneqq (x_0^2 + x^2)(y_0^2 + y^2)(z_0^2 + z^2) + Axyzx_0y_0z_0 - 2x_0^2y_0^2z_0^2.$$

Let $X_A \subset (\mathbb{P}^1)^3$ denote the hypersurface defined by the vanishing set of $\tilde{q}_A$. Then X_A is a smooth real projective surface (see Lemma B.1). A computation using the adjunction formula reveals the canonical bundle of X_A to be trivial. Moreover, the Lefschetz hyperplane theorem implies $b_1(X_A) = 0$. It follows that X_A is a K3 surface.

Each X_A is an example of a so-called $(2, 2, 2)$ surface in $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$, since q_A has degree two in each variable. There are three natural involutions of any $(2, 2, 2)$-surface which we recall here. Let $\pi_i : (\mathbb{P}^1)^3 \to \mathbb{P}^1$ denote projection to the ith coordinate. For $i \neq j$, the map $\pi_{ij} \coloneqq (\pi_i, \pi_j) : (\mathbb{P}^1)^3 \to (\mathbb{P}^1)^2$ restricted to X_A has fibers generically containing exactly two points and otherwise containing one point or a rational curve. We define σ_k^A to be the involution of X_A which preserves each fiber of π_{ij} $(i, j \neq k)$ and swaps elements of a generic fiber. Since any birational self-map of a K3 surface extends to an

automorphism, σ_k^A is an automorphism. The three involutions σ_i on any $(2,2,2)$-surface generate a free product $\mathbb{Z}/2\mathbb{Z} \star \mathbb{Z}/2\mathbb{Z} \star \mathbb{Z}/2\mathbb{Z}$ in $Aut(X)$ (see [20]).

For each nonzero $A \in \mathbb{R}$, we consider the automorphism of X_A given by

$$f_A \coloneqq \sigma_3^A \circ \sigma_2^A \circ \sigma_1^A.$$

In Section 2, we prove that $h_{top}(f_A, X_A(\mathbb{C}))$ is positive and independent of A.

Remark 1.1. In [16], Moncet uses a type of continuity of topological entropy to prove that

$$\lim_{A \to 0} h_{top}(f_A, X_A(\mathbb{R})) = 0.$$

He applies a theorem of Cantat from [2] to obtain that $h_{top}(f_A, X_A(\mathbb{C}))$ is positive and independent of A, implying $\lim_{A\to 0} \rho(f_A) = 0$. Our computation of $h_{top}(f_A, X_A(\mathbb{C}))$ differs from Moncet's; he assumes that the fibers of π_{ij} do not contain any rational curves, which is not the case. Regardless, $h_{top}(f_A, X_A(\mathbb{C}))$ remains positive and independent of A, so Moncet's conclusion holds.

Remark 1.2. We note that the family of examples $\{X_A\}$ is non-generic within all $(2,2,2)$-surfaces. Cantat proves in [2] that a generic $(2,2,2)$-surface has Picard number 3, while we will see that each X_A has Picard number at least 12. Moreover, Cantat proves that the automorphism $\sigma_3 \circ \sigma_2 \circ \sigma_1$ on a generic $(2,2,2)$-surface has entropy $\ln\left(9 + 4\sqrt{5}\right) \approx \ln 17.9443$, while we will find $h_{top}(f_A) \approx \ln 6.1393$.

While the calculations in Section 2 apply for any nonzero A, in Section 3 we focus on the specific case $A = 10$. The real surface $X_{10}(\mathbb{R})$ is homeomorphic to a sphere and completely contained in the affine part of X_{10} (see Figure 1). Define

$$Q(t) \coloneqq t^6 - 5t^5 - 6t^4 - 5t^3 - 6t^2 - 5t + 1, \tag{1.3}$$

and

$$\begin{aligned} P(t) \coloneqq & t^{14} - 16t^{13} + 78t^{12} - 108t^{11} - 99t^{10} + 334t^9 + 56t^8 \\ & - 556t^7 + 56t^6 + 334t^5 - 99t^4 - 108t^3 + 78t^2 - 16t + 1. \end{aligned} \tag{1.4}$$

Let μ be the unique real root of Q greater than 1, and let λ be the unique real root of P on the interval $(8.19981448, 8.19981449)$.

Theorem 1.5 (Main Theorem)**.** *For X_A, f_A, λ, and μ defined above:*

1. *$h_{top}(f_A, X_A(\mathbb{C})) = \ln \mu$ for every nonzero $A \in \mathbb{R}$;*
2. *there exists a finite f_{10}-invariant set $S \subset X_{10}(\mathbb{R})$ such that the mapping class of f_{10}^2 on $X_{10}(\mathbb{R}) \backslash S$ is pseudo-Anosov with stretching factor λ.*

As a consequence of (1) and (2),

$$\rho(f_{10}) \geq \frac{\ln \lambda}{2 \ln \mu} > \frac{1}{2}.$$

1.2. **Outline.** Our computation of $h_{top}(f_A, X_A(\mathbb{C}))$ in Section 2 relies on the Gromov–Yomdin theorem and working with explicit algebraic curves. In Section 3.2, we prove a checkable shadowing lemma for an arbitrary C^2-diffeomorphism of a (real) surface. In Sections 3.4 and 3.5, we apply the shadowing lemma to a pseudo-orbit of f_{10} on $X_{10}(\mathbb{R})$, with the help of a computer. The result is an f_{10}-periodic point $x \in X_{10}(\mathbb{R})$ with period 10.

In Section 3.6, we compute the mapping class of f_{10}^2 on $X_{10}(\mathbb{R}) \setminus \left\{ f^i(x) \mid i \in \mathbb{Z} \right\}$ in terms of Dehn half-twists. The mapping class representation of f_{10}^2 is written explicitly in Section 3.6.2. From there, Flipper ([1]) is able to compute the stretch factor (dilatation) of f_{10}^2, yielding a lower bound on $h_{top}(f_{10}, X_{10}(\mathbb{R}))$ (see [7]).

1.3. **Accompanying computer code.** The GitHub repository accompanying this article is K3entropy. See its `README` file for more details. All code is based on Curt McMullen's 'Orbits of Automorphisms of K3 Surfaces' (see [15]). Throughout, when we say that a program *certifies* a statement, we mean that it proves the statement using exact rational arithmetic or outward-rounded interval arithmetic. The outward-rounding procedure, implemented using the GNU MPFR library (see [8]), is described in Appendix C. Aside from `Flipper`, the three programs essential to the main theorem are:

- `verifyPeriodicOrbit.c` – utilized in Section 3.5 – certifies an upper bound for $\max_i \left\| f_i^c(0) \right\|_2$ (see Lemma 3.2 for notation).
- `derivativesInterval.c` – utilized in Sections 3.5.1 and 3.5.2 – certifies that the matrices $\widetilde{L}_i$ displayed in Table 4 satisfy $\left\| Df_i^c(0) - \widetilde{L}_i \right\|_\infty < 10^{-2}$. It also certifies bounds for $\mathcal{D}$ (see Section 3.4.1 for notation) and its first and second partials along a specific pseudo-orbit (Table 3).

- `verifyArcs.c` – used to produce the top panel of Figure 4 – certifies the relative homotopy classes of the images of several arcs under f_{10}.

In Section 3.6.1, we provide an algorithm for realizing the mapping class of a homeomorphism of an n-punctured sphere as a product of Dehn half-twists. The algorithm is implemented in `mclass.c` and takes as input the arc-data obtained by `verifyArcs.c`, stored in the file `arc_certificate_data.c`. The output of `mclass.c` may be verified by hand (see Figure 4).

Finally, we use Flipper to compute the stretch factor of the product of Dehn half-twists representing the mapping class of f_{10}^2 on $X_{10}(\mathbb{R}) \setminus \left\{f^i(x) \mid i \in \mathbb{Z}\right\}$. The file `FlipperDilatation.ipynb` is a Jupyter Notebook which runs Flipper 0.15.6 in SageMath 10.5 ([19]), with `realalg` 0.3.6. The notebook verifies that the mapping class is pseudo-Anosov, constructs the 42×42 hitting matrix for its invariant train track, verifies that P is the minimal polynomial of its stretching factor, and provides an interval containing the stretching factor.

1.4. **Notation.** Here we collect some notation used in the rest of the paper:

- For $v = (v_1, \dots, v_d) \in \mathbb{R}^d$, let $\|v\|_2 \coloneqq \left(\sum_{j=1}^d |v_j|^2\right)^{\frac{1}{2}}$.
- For $v = (v_1, \dots, v_d) \in \mathbb{R}^d$, let $\|v\|_\infty \coloneqq \max_{1\le j\le d} |v_j|$. We use the same notation for any matrix to denote its maximum absolute entry.
- $\|\cdot\|_0$ is the norm on $(\mathbb{R}^2)^n$ defined by $\|(\vec{v}_1, \dots, \vec{v}_n)\|_0 \coloneqq \max_{1\le i\le n} \|\vec{v}_i\|_2$.
- For $M \in M_{m\times n}(\mathbb{R})$, $\|M\|_{op}$ denotes the operator norm for M as a map $(\mathbb{R}^n, \|\cdot\|_2) \to (\mathbb{R}^m, \|\cdot\|_2)$.
- Given $M \in M_{2n\times 2n}(\mathbb{R})$, $\|M\|_{op,0}$ denotes the operator norm of M as a map $((\mathbb{R}^2)^n, \|\cdot\|_0) \to ((\mathbb{R}^2)^n, \|\cdot\|_0)$.
- For $M \in M_{m\times n}(\mathbb{R})$, $\|M\|_F$ is the Frobenius norm.
- For twice-differentiable $f : \mathbb{R}^n \to \mathbb{R}^m$, $D^2 f$ is the Hessian matrix.
- For $f : \mathbb{R}^n \to \mathbb{R}^m$, let $\|f\|_{C^0(U)} \coloneqq \sup_{x\in U} \|f(x)\|_2$.
- For $f \in C^k(\mathbb{R}^n)$, let $\|f\|_{C^k(U),\infty} \coloneqq \max_{|\beta|\le k} \sup_{x\in U} |D^\beta f(x)|$.
- For a C^k-function $f = (f_1, \dots, f_m) : \mathbb{R}^n \to \mathbb{R}^m$, let $\|f\|_{C^k(U),\infty} \coloneqq \max_i \|f_i\|_{C^k(U),\infty}$.
- For $G : \mathbb{R}^n \to M_{\ell\times k}(\mathbb{R})$, let $\|G\|_{U,\star} \coloneqq \sup_{x\in U} \|G(x)\|_\star$, where $\star \in \{op, F, \infty\}$.

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2. Computing $h_{top}(f, X(\mathbb{C}))$

Notation 2.1. For the rest of this section, we fix nonzero $A \in \mathbb{R}$ and let $X \coloneqq X_A$ and $f \coloneqq f_A$.

2.1. **Background.** The Gromov–Yomdin theorem implies that $h_{top}(f, X(\mathbb{C})) = \ln R(f^*)$, where $R(f^*)$ denotes the spectral radius of $f^* \in GL(H^*(X, \mathbb{C}))$. In this section, we recall a few relevant facts about K3 surfaces (see, for example, [11]). By Hodge theory, X satisfies

$$\dim H^k(X, \mathbb{C}) = \begin{cases} 1 & k = 0, 4 \\ 22 & k = 2 \\ 0 & \text{otherwise.} \end{cases}$$

Therefore, f^* achieves its spectral radius on $H^2(X, \mathbb{C})$. Additionally, there exists an $Aut(X)$-invariant decomposition of $H^2(X, \mathbb{C})$ into its Dolbeault cohomology classes, denoted

$$H^2(X, \mathbb{C}) = H^{2,0}(X, \mathbb{C}) \oplus H^{1,1}(X, \mathbb{C}) \oplus H^{0,2}(X, \mathbb{C}).$$

Complex conjugation preserves $H^{1,1}(X, \mathbb{C})$ and $H^{2,0}(X, \mathbb{C}) \oplus H^{0,2}(X, \mathbb{C})$, so both are spanned by elements of $H^2(X, \mathbb{R})$. By the Hodge Index Theorem, the intersection form $\langle \alpha, \beta \rangle \coloneqq \int_X \alpha \wedge \beta$ is positive-definite on $(H^{2,0}(X, \mathbb{C}) \oplus H^{0,2}(X, \mathbb{C})) \cap H^2(X, \mathbb{R})$ and has signature $(1, 19)$ on $H^{1,1}(X, \mathbb{R}) \coloneqq H^{1,1}(X, \mathbb{C}) \cap H^2(X, \mathbb{R})$. Consequently, f^* achieves its spectral radius on $H^{1,1}(X, \mathbb{R})$. Define the Néron-Severi group by

$$NS(X) \coloneqq H^2(X,\ \mathbb{Z}) \cap H^{1,1}(X,\ \mathbb{R})$$

in the sense that we consider the image of $H^2(X,\ \mathbb{Z})$ in $H^2(X,\ \mathbb{R})$ and intersect with $H^{1,1}(X,\ \mathbb{R})$. Then $NS(X)$ is a discrete subgroup of $H^2(X, \mathbb{R})$. Since $Aut(X)$ preserves $H^2(X, \mathbb{Z})$ and the Hodge decomposition, it preserves $NS(X)$. Since X is a K3 surface, there exist $Aut(X)$-equivariant isomorphisms

$$D(X) \cong Pic(X) \cong NS(X),$$

where $Pic(X)$ is the Picard group and $D(X)$ is the free abelian group of algebraic curves considered up to linear equivalence. Moreover, under these isomorphisms, $\langle\cdot,\cdot\rangle$ is carried to the intrinsic intersection forms on $D(X)$ and $Pic(X)$. Since X is projective, the Picard group contains a line bundle of positive self-intersection. Therefore, $\langle\cdot,\cdot\rangle$ has signature $(1,\rho(X)-1)$ on each of the real vector spaces

$$D(X)\otimes_{\mathbb{Z}}\mathbb{R}, \qquad Pic(X)\otimes_{\mathbb{Z}}\mathbb{R}, \qquad NS(X)\otimes_{\mathbb{Z}}\mathbb{R},$$

where $\rho(X) \coloneqq \operatorname{rank} Pic(X) = \operatorname{rank} NS(X)$ is the Picard number.

2.2. **Strategy.** We will find an explicit σ_i-invariant subspace W of $D(X)\otimes_{\mathbb{Z}}\mathbb{R}$ on which $\langle\cdot,\cdot\rangle$ is Minkowski. Then $R(f^*) = R(f^*\big|_W)$ will be directly computed.

2.3. **Computations on homology.** To that end, we consider several algebraic curves in X. In homogeneous coordinates $([x_0:x],[y_0:y],[z_0:z])$ for $\mathbb{P}^1\times\mathbb{P}^1\times\mathbb{P}^1$, let

$$\begin{aligned}
p_1 &\coloneqq \mathbb{P}^1\times\{[0:1]\}\times\{[1:i]\}, & p_2 &\coloneqq \mathbb{P}^1\times\{[0:1]\}\times\{[1:-i]\},\\
p_3 &\coloneqq \mathbb{P}^1\times\{[1:i]\}\times\{[0:1]\}, & p_4 &\coloneqq \mathbb{P}^1\times\{[1:-i]\}\times\{[0:1]\},\\
p_5 &\coloneqq \{[0:1]\}\times\mathbb{P}^1\times\{[1:i]\}, & p_6 &\coloneqq \{[0:1]\}\times\mathbb{P}^1\times\{[1:-i]\},\\
p_7 &\coloneqq \{[1:i]\}\times\mathbb{P}^1\times\{[0:1]\}, & p_8 &\coloneqq \{[1:-i]\}\times\mathbb{P}^1\times\{[0:1]\},\\
p_9 &\coloneqq \{[0:1]\}\times\{[1:i]\}\times\mathbb{P}^1, & p_{10} &\coloneqq \{[0:1]\}\times\{[1:-i]\}\times\mathbb{P}^1,\\
p_{11} &\coloneqq \{[1:i]\}\times\{[0:1]\}\times\mathbb{P}^1, & p_{12} &\coloneqq \{[1:-i]\}\times\{[0:1]\}\times\mathbb{P}^1.
\end{aligned}$$

Each p_i is birationally equivalent to $\mathbb{P}^1$, and thus has self-intersection -2 by the adjunction formula. Moreover, each p_i is the unique representative of $[p_i]\in D(X)$, so we use p_i and $[p_i]$ interchangeably. Let

$$W \coloneqq \operatorname{span}_{\mathbb{R}}\{p_1,\dots,p_{12}\}\subset D(X)\otimes_{\mathbb{Z}}\mathbb{R}.$$

A calculation, also performed in [18], yields

$$M \coloneqq (\langle p_i,p_j\rangle)_i^j = \begin{pmatrix}
-2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1\\
0 & -2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1\\
0 & 0 & -2 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0\\
0 & 0 & 0 & -2 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0\\
1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 1 & 1 & 0 & 0\\
0 & 1 & 0 & 0 & 0 & -2 & 0 & 0 & 1 & 1 & 0 & 0\\
0 & 0 & 1 & 1 & 0 & 0 & -2 & 0 & 0 & 0 & 1 & 0\\
0 & 0 & 1 & 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 1\\
0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & -2 & 0 & 0 & 0\\
0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & -2 & 0 & 0\\
1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -2 & 0\\
1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -2
\end{pmatrix},$$

and

$$S \coloneqq (\langle p_i,\sigma_1 p_j\rangle)_i^j = \begin{pmatrix}
-2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1\\
0 & -2 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1\\
0 & 0 & -2 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0\\
0 & 0 & 0 & -2 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0\\
1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 1 & 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 1\\
0 & 0 & 1 & 1 & 0 & 0 & -2 & 0 & 0 & 0 & 1 & 0\\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0\\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\
1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -2\\
1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -2 & 0
\end{pmatrix}.$$

The matrix M is straightforward to compute. For example, the curves p_1 and p_5 meet transversely at $([0:1],[0:1],[1:i])$, and thus $\langle p_1,p_5\rangle = 1$. To illustrate how to compute S, let's calculate $\langle p_1,\sigma_1(p_5)\rangle$. When $z=[1:i]$, the defining equation becomes $\tilde{q}_A([x_0:x],[y_0:y],[1:i]) = x_0y_0(Aixy - 2x_0y_0)$. For a fixed generic $[y_0:y]$, we have either $x_0=0$ or $Aixy = 2x_0y_0$. Since $x_0=0$ on p_5, it follows that $\sigma_1(p_5) = \{([x_0:x],[y_0:y],[1:i]) \mid Aixy = 2x_0y_0\}$. This curve meets p_1 transversely at $([1:0],[0:1],[1:i])$, so $\langle p_1,\sigma_1(p_5)\rangle = 1$.

The matrix M is invertible, so $p_1, \dots, p_{12}$ are linearly independent. Diagonalizing M shows that $\langle \cdot, \cdot \rangle$ has signature $(1, 11)$ on W. Alternatively, we can realize an element of W with positive self-intersection in the following way. Each fiber of $\pi_i : X_A \to \mathbb{P}^1$ is a curve given by the equation $\{x_i = \alpha\}$, all representing the same class in $D(X)$, which we denote by $[c_i]$. Notice $\{x = [0:1]\} = p_5 \cup p_6 \cup p_9 \cup p_{10}$ so $[c_1] = p_5 + p_6 + p_9 + p_{10}$. Similarly, $[c_2] = p_1 + p_2 + p_{11} + p_{12}$ and $[c_3] = p_3 + p_4 + p_7 + p_8$. So each $[c_i]$ is contained in W. One can also check $\langle [c_i], [c_j] \rangle = 2$ whenever $i \neq j$ and vanishes otherwise. Thus $[c_i] + [c_j]$, $i \neq j$, has positive self-intersection.

Define $\tilde{\sigma}_1 : W \to W$ by $\tilde{\sigma}_1 \coloneqq M^{-1}S$, so that $\tilde{\sigma}_1$ has the property

$$\langle \sigma_1^* w_1, w_2 \rangle = \langle \tilde{\sigma}_1 w_1, w_2 \rangle, \tag{2.2}$$

for all $w_1, w_2 \in W$. One finds

$$\tilde{\sigma}_1 = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

The fact that $\tilde{\sigma}_1$ has integer entries and satisfies $\tilde{\sigma}_1^2 = id$ was not a priori guaranteed and suggests the following claim.

Claim 2.3. $\tilde{\sigma}_1 = \sigma_1^*\big|_W$

Proof. Notice that $\tilde{\sigma}_1^t M \tilde{\sigma}_1 = SM^{-1}S = M$. Therefore $\langle \cdot, \cdot \rangle$ is $\tilde{\sigma}_1$-invariant. Now σ_1^* is induced by an automorphism of X and thus also preserves $\langle \cdot, \cdot \rangle$.

Fix $w \in W$ and set $d \coloneqq \sigma_1^*(w) - \tilde{\sigma}_1(w)$. For every $u \in W$, equation (2.2) gives

$$\langle d, u \rangle = \langle \sigma_1^*(w), u \rangle - \langle \tilde{\sigma}_1(w), u \rangle = 0.$$

Thus $d \in W^\perp$. Moreover, applying (2.2) with $u = \tilde{\sigma}_1(w)$ and using the invariance of the intersection form gives

$$\begin{aligned} \langle d, d \rangle &= \langle \sigma_1^*(w), \sigma_1^*(w) \rangle + \langle \tilde{\sigma}_1(w), \tilde{\sigma}_1(w) \rangle - 2\langle \sigma_1^*(w), \tilde{\sigma}_1(w) \rangle \\ &= \langle w, w \rangle + \langle w, w \rangle - 2\langle \tilde{\sigma}_1(w), \tilde{\sigma}_1(w) \rangle = 0. \end{aligned}$$

Since $\langle \cdot, \cdot \rangle$ is negative definite on $W^\perp$, it follows that $d = 0$. Therefore $\sigma_1^*(w) = \tilde{\sigma}_1(w)$ for every $w \in W$, proving the claim. □

Let's check that σ_1^* does what we expect to $[c_1], [c_2], [c_3]$. Well, σ_1^* should fix $[c_2] = p_1 + p_2 + p_{11} + p_{12}$ and $[c_3] = p_3 + p_4 + p_7 + p_8$; our matrix does this. Moreover,

$$\pi_{23}^{-1}(\pi_{23}(c_1)) = \sigma_1(c_1) \cup c_1 \cup p_1 \cup p_2 \cup p_3 \cup p_4. \tag{2.4}$$

Now $\pi_{23}(c_1) \subset \mathbb{P}^1 \times \mathbb{P}^1$ has bidegree $(2, 2)$ and therefore represents the class $2[\{\alpha\} \times \mathbb{P}^1] + 2[\mathbb{P}^1 \times \{\alpha\}]$. Thus (2.4) becomes

$$2[c_2] + 2[c_3] = \sigma_1^*[c_1] + [c_1] + p_1 + p_2 + p_3 + p_4,$$

and after expansion

$$\sigma_1^*(p_5 + p_6 + p_9 + p_{10}) = 2(p_{11} + p_{12} + p_7 + p_8) + p_1 + p_2 + p_3 + p_4 - p_5 - p_6 - p_9 - p_{10},$$

which our matrix also satisfies. A similar computation is performed in [17].

Let P_{xy} denote the permutation matrix corresponding to

$$(1, 5)(2, 6)(3, 7)(4, 8)(9, 11)(10, 12),$$

and let P_{xz} denote the permutation matrix corresponding to

$$(1, 11)(2, 12)(3, 9)(4, 10)(5, 7)(6, 8).$$

Using symmetry, one obtains

$$\sigma_2^*\big|_W = P_{xy}\tilde{\sigma}_1 P_{xy}^{-1} = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}$$

and

$$\sigma_3^*\big|_W = P_{xz}\tilde{\sigma}_1 P_{xz}^{-1} = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Thus

$$f^*\big|_W = \sigma_1^*\big|_W \sigma_2^*\big|_W \sigma_3^*\big|_W = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 & -1 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & -1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 2 & 1 & 1 & 0 & 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 2 & 2 & 1 & 1 & 0 & 0 & -1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 1 & 0 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 3 & 3 & 1 & 1 & 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 3 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & -2 & -1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & -1 & -2 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & -1 \\ 0 & 0 & 2 & 2 & 0 & 0 & 0 & 1 & 1 & 1 & 2 & 1 \\ 0 & 0 & 2 & 2 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 2 \end{pmatrix}.$$

Since $\langle \cdot, \cdot \rangle$ has signature $(1, 11)$ on W and signature $(1, 19)$ on $H^{1,1}(X, \mathbb{R})$, it is negative definite on the orthogonal complement $W^\perp \subset H^{1,1}(X, \mathbb{R})$. Because W is f^*-invariant and f^* preserves the intersection form, $W^\perp$ is also f^*-invariant. Thus $f^*\big|_{W^\perp}$ is an isometry of a negative-definite space and has spectral radius 1. Hence, $R(f^*) = R\left(f^*\big|_W\right)$. We find that the characteristic polynomial of $f^*\big|_W$ is

$$r(x) = Q(x)(x^2 - x + 1)^3,$$

where Q is as defined earlier. The factor $x^2 - x + 1$ is cyclotomic, while Q is a Salem polynomial. In particular, the spectral radius $R(f^*\big|_W)$ is the unique real root of Q of modulus > 1, namely $\mu \approx 6.1393$. By the Gromov–Yomdin theorem,

$$h_{top}(f, X(\mathbb{C})) = \ln \mu \approx \ln 6.1393. \tag{2.5}$$

Remark 2.6. This calculation demonstrates that $h_{top}(f_A)$ is independent of $A \in \mathbb{R}^*$.

3. Bounding $h_{top}(f, X_{10}(\mathbb{R}))$

Notation 3.1. For the rest of the article, we let $X := X_{10}$, $f := f_{10}$, $q := q_{10}$, and $\tilde{q} := \tilde{q}_{10}$.

3.1. **Strategy.** Given a surface $S = S_{g,n}$, recall that the mapping class group of S is

$$\mathrm{MOD}^{\pm}(S) \coloneqq \mathrm{Homeo}(S)/\mathrm{Homeo}_0(S),$$

noting that we allow for orientation-reversing homeomorphisms. Let $\mathrm{MOD}^{+}(S)$ denote the index-2 subgroup of $\mathrm{MOD}^{\pm}(S)$ consisting of classes coming from orientation-preserving homeomorphisms. Given a homeomorphism h of S, we let $[h] \in \mathrm{MOD}^{\pm}(S)$ denote its mapping class.

In the following sections, we will approximate an f-periodic point $x \in X(\mathbb{R})$ of order 10. Then we examine the action of f on $S_{0,10} \cong X(\mathbb{R}) \setminus \left\{f^i(x)\right\}_{i=0}^{9}$. Our goal is to represent $[f^2]$ as a product of Dehn half-twists in $\mathrm{MOD}^{+}(S_{0,10})$. Such a representation allows Flipper to show that $[f^2]$ is a pseudo-Anosov mapping class and compute its stretch factor, denoted $\lambda([f^2])$. Since each σ_i is orientation-reversing on $X(\mathbb{R})$, f is orientation-reversing, which is why we consider f^2. Recall that a pseudo-Anosov homeomorphism φ has minimal entropy in its mapping class, and moreover $h_{top}(\varphi) = \ln \lambda([\varphi])$ (see [7]). Therefore,

$$h_{top}(f) = \frac{1}{2} h_{top}(f^2) \geq \frac{1}{2} \ln \lambda([f^2]).$$

Now, we turn to approximating such a periodic point x. Since f has positive entropy on $X(\mathbb{C})$, a classical theorem of Katok implies that the number of n-periodic points in $X(\mathbb{C})$ grows at least exponentially in n (see [13]). However, finding explicit periodic points remains a challenge, especially those in $X(\mathbb{R})$. To that end, we prove an ad hoc shadowing lemma for C^2 surface diffeomorphisms, and then use a computer to exhibit a pseudo-orbit (x_i) with sufficient recurrence and hyperbolicity to admit a nearby periodic point.

3.2. **A shadowing lemma.** The following shadowing lemma can be thought of as an ad hoc version of the Anosov shadowing lemma, where we only care about data along a given pseudo-orbit. The lemma requires some type of hyperbolicity in coordinates along the pseudo-orbit, control of the second derivative in coordinates along the pseudo-orbit, and high recurrence. The benefit is that all constants can be computed explicitly.

Lemma 3.2. *Let M be a smooth surface. Let*

- *$h : M \to M$ be a C^2 diffeomorphism*
- *$x_0, \dots, x_{n-1} \in M$*
- *$\{\varphi_i : V_i \to M\}$ be a set of C^2 charts such that $x_i = \varphi_i(0)$.*

Assume $h(x_i) \in \varphi_{i+1 \bmod n}(V_{i+1 \bmod n})$ for all $0 \leq i < n$. For each such i, set

- *$U_i \coloneqq \{v \in V_i \mid h(\varphi_i(v)) \in \varphi_{i+1 \bmod n}(V_{i+1 \bmod n})\}$*
- *$h_i^c \coloneqq \varphi_{i+1 \bmod n}^{-1} \circ h \circ \varphi_i : U_i \to \mathbb{R}^2$*
- *$L_i \coloneqq (Dh_i^c)_0$.*

Define $L : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ by

$$L \coloneqq \begin{pmatrix} 0 & 0 & \cdots & 0 & L_{n-1} \\ L_0 & 0 & \cdots & 0 & 0 \\ 0 & L_1 & \ddots & \vdots & \vdots \\ \vdots & & \ddots & 0 & 0 \\ 0 & \cdots & 0 & L_{n-2} & 0 \end{pmatrix}.$$

Assuming $L - I_{2n}$ is invertible (or equivalently $L_{n-1}L_{n-2}\cdots L_0 - I_2$ is invertible), let $\kappa \coloneqq \left\|(L - I_{2n})^{-1}\right\|_{op,0}$ (recalling Notation 1.4). If there exists $\delta > 0$ such that

(1) for all i, $B_{12\kappa\delta}(0) \subset U_i$
(2) for all i,

$$12\kappa\delta < \frac{1}{\kappa} \frac{1}{16\|D^2 h_i^c\|_{C^0(B_{12\kappa\delta}(0)),\infty}}$$

(3) for all i, $\|h_i^c(0)\|_2 < \delta$

then there exists an h-periodic point $p \in M$ satisfying $h^n(p) = p$ and $h^i(p) \in \varphi_i(B_{6\kappa\delta}(0))$.

Our proof mirrors Katok and Hasselblatt's proof of the Anosov shadowing lemma in [14], Theorem 6.4.15. We delay it until Appendix A.

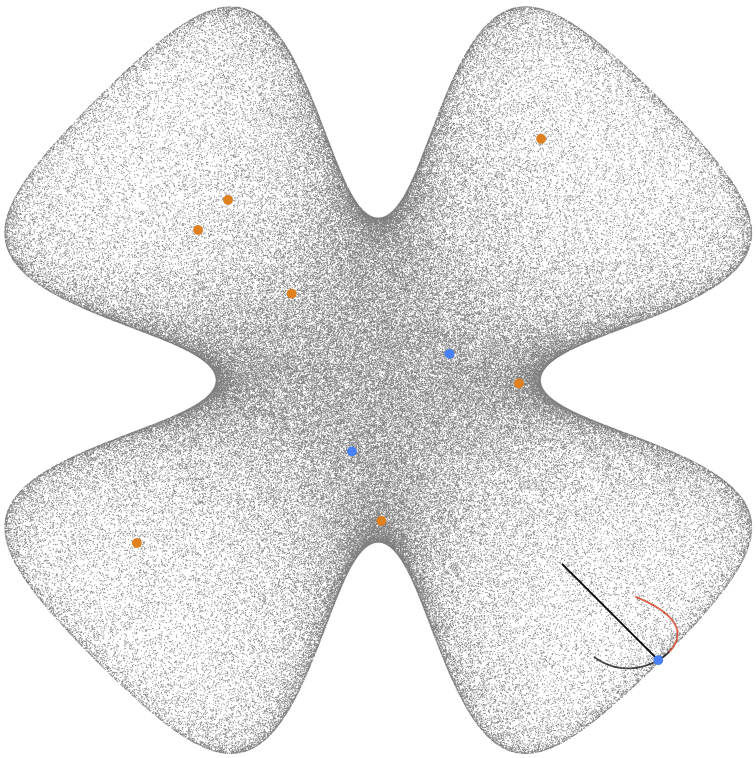


FIGURE 2. The pseudo-orbit $x_i = \tilde{\varphi}_i(a_i, b_i)$ from Table 1 plotted on $X(\mathbb{R})$, and projected into the z-x plane. Blue dots denote points on the top sheet and orange dots denote points on the bottom sheet. The curve $C \coloneqq \{x = -z\} \cap X$ can be seen intersecting $f^5(C)$ near x_0. This figure is obtained using a program developed by C. McMullen.

3.3. **Where to look for pseudo-orbits.** Let $\rho : (\mathbb{P}^1)^3 \to (\mathbb{P}^1)^3$ denote reflection through $\{x = -z\}$. Notice that $\tilde{q}$ is ρ-invariant, so $\rho(X) = X$. Furthermore, $C \coloneqq \{x = -z\} \cap X$ is cut out by an equation of bidegree $(4, 2)$ in $\mathbb{P}^1 \times \mathbb{P}^1$. Consequently, C has arithmetic genus $3 = (4-1)(2-1)$ and thus self-intersection $4 = 3 \cdot 2 - 2$ by the adjunction formula for a K3 surface. A simple computation reveals that $\rho \circ f = f^{-1} \circ \rho$. Therefore, any $p \in C \cap f^n(C)$ is f-periodic of order dividing $2n$.

Recall that there is a unique projective class in the isotropic cone of $\langle \cdot, \cdot \rangle$ in $H^{1,1}(X, \mathbb{R})$ that is the attracting fixed point for the action of f. Let v_f^+ denote a representative of this class lying in the closure of the component of the positive cone containing $[C]$. Similarly, f has a repelling projective class from which we choose a representative v_f^- in the closure of the same component, normalized so that $\langle v_f^+, v_f^- \rangle = 1$. Then

$$\lim_{n \to \infty} e^{-nh_{top}(f)} \langle C, f^n(C) \rangle = \langle C, v_f^- \rangle \cdot \langle C, v_f^+ \rangle$$

(see [3], section 2.3). Since $[C]$ has positive self-intersection and $v_f^\pm$ lie in the closure of the same positive-cone component as $[C]$, $\langle C, v_f^- \rangle$ and $\langle C, v_f^+ \rangle$ are positive. Thus, the intersection number $\langle C, f^n(C) \rangle$ grows like $e^{nh_{top}(f)}$. Since each point of $C \cap f^n(C)$ is fixed by f^{2n}, these intersections provide a natural place to search for periodic points. A priori, these intersection points need not be contained in $X(\mathbb{R})$. However, one can plot C and $f^n(C)$ and see many intersections in $X(\mathbb{R})$; in the following section, we will input into Lemma 3.2 a pseudo-orbit coming from an approximate intersection point of C and $f^5(C)$ in $X(\mathbb{R})$.

It's worth noting that, if proj_W denotes the orthogonal projection onto W with respect to the intersection form, one can calculate the intersection of C with each p_i and use the intersection matrix from Section 2.3 to arrive at $\langle \mathrm{proj}_W[C], \mathrm{proj}_W[C] \rangle = 4$. But $\langle [C], [C] \rangle = 4$ and $W^\perp$ is negative-definite, so in fact $[C] \in W$. Therefore, our formula for the action of f^* on W allows us to explicitly compute $\langle C, f^n(C) \rangle$ for any given n.

3.4. **A specific pseudo-orbit.** By the previous section, any intersection between C and $f^n(C)$ in $X(\mathbb{R})$ has period dividing $2n$. Figure 2 shows such an intersection when $n = 5$. In the figure, we selected a point near the intersection and plotted the first ten points in its orbit. This will serve as our pseudo-orbit. In the following section, we define the φ_i used in Lemma 3.2.

3.4.1. *Charts.* Let $\mathcal{D}(x,y)$ denote the discriminant of q with respect to z, i.e.,

$$\mathcal{D}(x,y) \coloneqq 10^2x^2y^2 + 8(1+x^2)(1+y^2) - 4(1+x^2)^2(1+y^2)^2.$$

Let

$$\Omega \coloneqq \left\{(u,v) \in \mathbb{C}^2 \mid u,v \notin \{\pm i\} \text{ and } \mathcal{D}(u,v) \notin (-\infty, 0]\right\}.$$

On Ω, define

$$p_\pm(u,v) \coloneqq \frac{-10uv \pm \mathcal{D}(u,v)^{\frac{1}{2}}}{2(1+u^2)(1+v^2)},$$

where $w \mapsto w^{\frac{1}{2}}$ denotes the principal branch of the square root, holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. Then the maps

$$\Psi_1^\pm(y,z) \coloneqq ([1 : p_\pm(y,z)], [1 : y], [1 : z]), \qquad \Psi_2^\pm(x,z) \coloneqq ([1 : x], [1 : p_\pm(x,z)], [1 : z]),$$
$$\Psi_3^\pm(x,y) \coloneqq ([1 : x], [1 : y], [1 : p_\pm(x,y)])$$

are inverse holomorphic coordinate charts on X defined on Ω. A direct calculation gives

$$(\partial_z q(x,y,z))^2 - \mathcal{D}(x,y) = 4(1+x^2)(1+y^2)q(x,y,z).$$

Hence, on $X = \{q = 0\}$, $\mathcal{D}(x,y) = (\partial_z q(x,y,z))^2$. By symmetry, analogous identities hold for the other two partial derivatives of q. Therefore, since X is smooth, $\mathcal{D}(x,y)$, $\mathcal{D}(x,z)$ and $\mathcal{D}(y,z)$ cannot vanish simultaneously on X. It follows that the restrictions of the three coordinate projections to the six open subsets $\Psi_j^\pm(\Omega \cap \mathbb{R}^2) \subset X(\mathbb{R})$, for $j = 1,2,3$, whose inverses are the corresponding maps $\Psi_j^\pm$, form an atlas of $X(\mathbb{R})$.

We consider the collection of points $(a_i, b_i) \in \mathbb{Q}^2$ and charts $\tilde{\varphi}_i$ enumerated in Table 1. For each i, define $x_i \coloneqq \tilde{\varphi}_i(a_i, b_i)$. Moreover, let

$$\varphi_i(\zeta, \gamma) \coloneqq \tilde{\varphi}_i(\zeta + a_i, \gamma + b_i),$$

so that the φ_i's and x_i's are as in Lemma 3.2. Here

$$V_i = \left\{(x,y) \in \mathbb{R}^2 \mid \mathcal{D}(x + a_i, y + b_i) > 0\right\}$$

and

$$U_i \coloneqq \left\{v \in V_i \mid f(\varphi_i(v)) \in \varphi_{i+1 \bmod 10}(V_{i+1 \bmod 10})\right\}.$$

Set

$$f_i^c \coloneqq \varphi_{i+1 \bmod 10}^{-1} \circ f \circ \varphi_i : U_i \to \mathbb{R}^2.$$

We emphasize that the *exact* values of the a_i and b_i are displayed in Table 1. We will see that this pseudo-orbit has enough recurrence and hyperbolicity to apply Lemma 3.2. One could choose a_i, b_i more carefully to get even more recurrence, with a negligible change in the first and second derivatives of f_i^c.

3.5. **Applying the shadowing lemma.** Recall the notation used in Lemma 3.2. Fix $\varepsilon' \coloneqq 10^{-18}$. In Section 3.5.1, we prove that $B_{\varepsilon'}(0) \subset U_i$ and $\max_i \left\|D^2 f_i^c\right\|_{C^0(B_{\varepsilon'}(0)),\infty} < 1.8 \cdot 10^{14}$. Moreover, in Section 3.5.2 we prove that $\kappa < 83$ for our choices of φ_i. In K3entropy, `verifyPeriodicOrbit.c` uses outward-rounded interval arithmetic based on the GNU MPFR library to certify that $0 \in U_i$ and $\|f_i^c(0)\|_2 < 10^{-29}$ for all i; see Appendix C. Set $\delta \coloneqq 10^{-29}$. Then

$$12\kappa\delta < 10^{-26} < \varepsilon',$$

and

$$12\kappa\delta < 10^{-26} < \frac{1}{16 \cdot 83 \cdot 1.8 \cdot 10^{14}} < \frac{1}{16\kappa \max_i \|D^2 f_i^c\|_{C^0(B_{\varepsilon'}(0)),\infty}}.$$

Therefore, Lemma 3.2 gives a point $x \in X(\mathbb{R})$ such that $f^{10}(x) = x$ and

$$\left\|\varphi_i^{-1}(f^i(x))\right\|_2 < \eta, \tag{3.3}$$

for each $0 \le i \le 9$, where $\eta \coloneqq 6\kappa\delta$. For the computer-assisted certifications below, set

$$\bar{\eta} \coloneqq 5 \cdot 10^{-27}.$$

Then $\eta < \bar{\eta}$, so $B_\eta(0) \subset [-\bar{\eta}, \bar{\eta}]^2$.

i	$\tilde{\varphi}_i$	a_i	b_i
0	Ψ_1^-	1.041643093944314148360673792017	1.726895448754858426328854724474
1	Ψ_3^-	-0.439586738044637984442175311821	0.555943953085459715621476373770
2	Ψ_2^-	1.111402054756051352317454938205	-0.926435350008842162121508383319
3	Ψ_1^-	-0.328869789067645570794396144391	0.867950394543647373540816310310
4	Ψ_2^-	1.488818954806569814700993326668	1.004464450964796276608907444033
5	Ψ_2^-	0.533829900932504729554816817729	-0.533829900932504729554816817729
6	Ψ_2^-	-1.004464450964796276608907444033	-1.488818954806569814700993326668
7	Ψ_3^+	-0.867950394543647373540816310310	-0.328869789067645570794396144391
8	Ψ_2^-	0.926435350008842162121508383319	-1.111402054756051352317454938205
9	Ψ_1^+	0.555943953085459715621476373770	0.439586738044637984442175311821

TABLE 1. A pseudo-orbit

3.5.1. *Bounding* $\left\|D^2 f_i^c\right\|_{C^0(B_\varepsilon(0)),\infty}$ *for small* ε. Define

$$K \coloneqq \max_i |\mathcal{D}(a_i, b_i)|, \qquad R \coloneqq \max_i \|D\mathcal{D}(a_i, b_i)\|_\infty, \qquad M \coloneqq \max_i \left\|D^2\mathcal{D}(a_i, b_i)\right\|_\infty.$$

In K3entropy, the program `derivativesInterval.c` uses outward-rounded interval arithmetic in the GNU MPFR library to enclose $\mathcal{D}(a_i, b_i)$ and its first and second partials at (a_i, b_i) for each i. The interval enclosures certify that the entries displayed in Table 3 are the correct roundings of these values to two decimals. It follows from Table 3 that $K < 115$, $R < 163$ and $M < 442$. The program `derivativesInterval.c` also certifies that $\mathcal{D} > 0$ on $B_{10^{-3}}(a_i, b_i)$ for all i; see Appendix C. In other words, $B_{10^{-3}}(0) \subset V_i$ for all i.

For each i, let $\xi_i : \mathbb{R}^3 \to \mathbb{R}^2$ be the affine map which agrees with φ_i^{-1} on $\varphi_i(V_i)$. In what follows, allowing an abuse of notation, we consider f_i^c to have domain V_i instead of U_i by setting

$$f_i^c \coloneqq \xi_{i+1 \bmod 10} \circ f \circ \varphi_i.$$

Lemma 3.4. *Fix* $\varepsilon \coloneqq 10^{-5}$. *Then* $\max_i \|Df_i^c\|_{C^1(B_\varepsilon(0)),\infty} < 1.8 \cdot 10^{14}$.

Proof of Lemma 3.4. In Lemma B.9 of the Appendix, we bound $\|Df\|_{C^0(X(\mathbb{R})),\infty}$ and $\left\|D^2 f\right\|_{C^0(X(\mathbb{R})),\infty}$, where f is considered to be a function from a neighborhood of $X(\mathbb{R}) \subset \mathbb{R}^3$ into $\mathbb{R}^3$. The remaining piece is to bound $\|Dp\|_{C^0(B_\varepsilon(a_i,b_i)),\infty}$ and $\left\|D^2 p\right\|_{C^0(B_\varepsilon(a_i,b_i)),\infty}$, and apply the chain rule. We proceed via a series of short claims.

Claim 3.5. *For all* i,

$$\max_{|\beta|=2} \sup_{B_\varepsilon(a_i,b_i)} |D^\beta \mathcal{D}| \le \sqrt{2} \cdot 20000 \cdot \varepsilon + M.$$

Claim 3.6. *For all* i,

$$\max_{|\beta|=1} \sup_{B_\varepsilon(a_i,b_i)} |D^\beta \mathcal{D}| \le \sqrt{2}(\sqrt{2} \cdot 20000 \cdot \varepsilon + M)\varepsilon + R.$$

Claim 3.7. *For all* i,

$$\sup_{B_\varepsilon(a_i,b_i)} |\mathcal{D}| \le \sqrt{2}(\sqrt{2}(\sqrt{2} \cdot 20000 \cdot \varepsilon + M)\varepsilon + R)\varepsilon + K.$$

Claim 3.5 follows from Lemma B.8 of the Appendix and the mean value theorem, with the $\sqrt{2}$ picked up as a dimensionality constant. Claim 3.6 similarly follows from Claim 3.5, and Claim 3.7 from Claim 3.6.

Claim 3.8. *For all* i,

$$\left\|D^2 p\right\|_{C^0(B_\varepsilon(a_i,b_i)),\infty} \le 2.5 \cdot 10^4$$

and

$$\|Dp\|_{C^0(B_\varepsilon(a_i,b_i)),\infty} < 122.$$

Since $\varepsilon = 10^{-5}$, Claim 3.5 gives

$$\max_i \max_{|\beta|=2} \sup_{B_\varepsilon(a_i,b_i)} |D^\beta \mathcal{D}| < \sqrt{2} \cdot 20000 \cdot 10^{-5} + 442 < 443.$$

Claim 3.6 then gives

$$\max_i \max_{|\beta|=1} \sup_{B_\varepsilon(a_i,b_i)} |D^\beta \mathcal{D}| < \sqrt{2} \cdot 443 \cdot 10^{-5} + 163 < 164,$$

so

$$\max_i \|D\mathcal{D}\|_{C^1(B_\varepsilon(a_i,b_i)),\infty} < 443.$$

Finally, Claim 3.7 gives

$$\max_i \sup_{B_\varepsilon(a_i,b_i)} |\mathcal{D}| < \sqrt{2} \cdot 164 \cdot 10^{-5} + 115 < 116.$$

Moreover, the same mean-value estimate and Table 3 give $\mathcal{D}(x,y) > 8$ for every $(x,y) \in B_\varepsilon(a_i,b_i)$ and every i. Then Lemma B.10 of the Appendix gives

$$\|Dp_\pm\|_{C^0(B_\varepsilon(a_i,b_i)),\infty} \leq 5 + \frac{1}{2}\sqrt{116} + \frac{1}{4} \cdot 443 < 122$$

and

$$\left\|D^2 p_\pm\right\|_{C^0(B_\varepsilon(a_i,b_i)),\infty} \leq 10 + \sqrt{116} + \frac{3}{4} \cdot 443 + \frac{1}{8} \cdot 443^2 < 2.5 \cdot 10^4,$$

proving Claim 3.8.

To conclude the proof of Lemma 3.4, we examine the action of f_i^c on the space of 2-jets from $\mathbb{R}^2$ to $\mathbb{R}$ (see Appendix B.1). Indeed, this action is given by a matrix-valued map $J^2(f_i^c) : B_\varepsilon(0) \to M_{5\times 5}(\mathbb{R})$ whose entries contain the first and second partials of f_i^c. Therefore we have the bound

$$\|Df_i^c\|_{C^1(B_\varepsilon(0)),\infty} \leq \left\|J^2(f_i^c)\right\|_{C^0(B_\varepsilon(0)),\infty}.$$

Jets behave nicely under composition, meaning

$$J_x^2(f_i^c) = J_x^2(\varphi_i) \cdot J_{\varphi_i(x)}^2(f) \cdot J_{f\circ\varphi_i(x)}^2(\xi_{i+1}).$$

For matrices A and B where A has r columns and B has r rows, the maximum-entry norm satisfies

$$\|AB\|_\infty \leq r\|A\|_\infty \|B\|_\infty.$$

Now $J_x^2(\varphi_i)$ is 5×9, $J_{\varphi_i(x)}^2(f)$ is 9×9, and $J_{f\circ\varphi_i(x)}^2(\xi_{i+1})$ is 9×5. Hence

$$\left\|J_x^2(f_i^c)\right\|_\infty \leq 9\left\|J_x^2(\varphi_i)\right\|_\infty \left\|J_{\varphi_i(x)}^2(f) \cdot J_{f\circ\varphi_i(x)}^2(\xi_{i+1})\right\|_\infty.$$

But ξ_{i+1} is an affine coordinate projection followed by a translation, so $J_{f\circ\varphi_i(x)}^2(\xi_{i+1})$ is an inclusion matrix, and

$$\left\|J_{\varphi_i(x)}^2(f) \cdot J_{f\circ\varphi_i(x)}^2(\xi_{i+1})\right\|_\infty \leq \left\|J_{\varphi_i(x)}^2(f)\right\|_\infty.$$

Moreover, by (B.11), $J^2(f)$ satisfies

$$\left\|J^2(f)\right\|_{X(\mathbb{R}),\infty} \leq \max\left(\|Df\|_{C^0(X(\mathbb{R})),\infty}, \left\|D^2 f\right\|_{C^0(X(\mathbb{R})),\infty}, 2\|Df\|_{C^0(X(\mathbb{R})),\infty}^2\right).$$

By Lemma B.9, $\|Df\|_{C^0(X(\mathbb{R})),\infty} \leq 9 \cdot 10^3$ and $\left\|D^2 f\right\|_{C^0(X(\mathbb{R})),\infty} \leq 9^2(2 \cdot 10^2)^3$. It follows that

$$\left\|J^2(f)\right\|_{X(\mathbb{R}),\infty} \leq 9^2(2 \cdot 10^2)^3.$$

Finally, $J^2(\varphi_i)$ satisfies

$$\left\|J^2(\varphi_i)\right\|_{C^0(B_\varepsilon(0)),\infty} \leq \max\left(1, \|Dp\|_{C^0(B_\varepsilon(a_i,b_i)),\infty}, \left\|D^2 p\right\|_{C^0(B_\varepsilon(a_i,b_i)),\infty}, 2\max\left(1, \|Dp\|_{C^0(B_\varepsilon(a_i,b_i)),\infty}\right)^2\right),$$

and by Claim 3.8, the right-hand side is bounded above by

$$\max(2 \cdot 122^2, 2.5 \cdot 10^4) < 3 \cdot 10^4.$$

Consequently,

$$\begin{aligned}\|Df_i^c\|_{C^1(B_\varepsilon(0)),\infty} \leq \left\|J^2(f_i^c)\right\|_{C^0(B_\varepsilon(0)),\infty} &\leq 9\left\|J^2(\varphi_i)\right\|_{C^0(B_\varepsilon(0)),\infty}\left\|J^2(f)\right\|_{X(\mathbb{R}),\infty} \\ &< 9 \cdot 3 \cdot 10^4 \cdot 9^2(2 \cdot 10^2)^3 \\ &< 1.8 \cdot 10^{14},\end{aligned}$$

proving Lemma 3.4. □

Lemma 3.9. *Fix $\varepsilon' := 10^{-18}$. Assume that $0 \in U_i$ for all i and that $\max_i \|f_i^c(0)\|_2 < \frac{1}{2}10^{-3}$. Then $B_{\varepsilon'}(0) \subset U_i$ for all i.*

Proof of Lemma 3.9. By Lemma 3.4, we have

$$\max_i \|Df_i^c\|_{C^0(B_{\varepsilon'}(0)),\infty} < 1.8 \cdot 10^{14}, \tag{3.10}$$

which implies $\max_i \|Df_i^c\|_{C^0(B_{\varepsilon'}(0)),op} < 2 \cdot 1.8 \cdot 10^{14}$. We chose ε' such that

$$\varepsilon' \cdot 2 \cdot 1.8 \cdot 10^{14} < \frac{1}{2} 10^{-3}.$$

Then for each i, the image of the ε'-ball under f_i^c satisfies

$$f_i^c(B_{\varepsilon'}(0)) \subset B_{\frac{1}{2}10^{-3}}(f_i^c(0)).$$

By assumption, $\|f_i^c(0)\|_2 < \frac{1}{2}10^{-3}$, so it follows that

$$f_i^c(B_{\varepsilon'}(0)) \subset B_{10^{-3}}(0) \subset V_{i+1 \bmod 10}.$$

Now

$$\xi_{i+1 \bmod 10}^{-1}\left(B_{\frac{1}{2}10^{-3}}(f_i^c(0))\right) \cap X(\mathbb{R})$$

has two connected components, one on each of the two sheets over $B_{\frac{1}{2}10^{-3}}(f_i^c(0))$. The connected set $f \circ \varphi_i(B_{\varepsilon'}(0))$ is contained in their union. Since $0 \in U_i$, this set meets the component corresponding to $\varphi_{i+1 \bmod 10}$, and hence is contained entirely in that component. Therefore, $B_{\varepsilon'}(0) \subset U_i$. □

3.5.2. *Bounding κ.* Recall the definitions of L_i, L from Lemma 3.2. We hope to bound

$$\kappa \coloneqq \left\|(L - I_{20})^{-1}\right\|_{op,0}.$$

Let $\widetilde{L}_i \in M_{2\times 2}(\mathbb{Q})$ be the matrices displayed in Table 4. Then the program `derivativesInterval.c` in K3entropy certifies that

$$\left\|L_i - \widetilde{L}_i\right\|_\infty < 10^{-2}.$$

Let $\widetilde{L}$ be the cyclic block matrix formed from the $\widetilde{L}_i$ which is the analog of L in Lemma 3.2. Write $(\widetilde{L} - I_{20})^{-1} = (B_{ij})_{0\le i,j<10}$ in 2×2 blocks. Since $\|B_{ij}\|_{op} \le \|B_{ij}\|_F$, the triangle inequality applied to each block row gives

$$\left\|(\widetilde{L} - I_{20})^{-1}\right\|_{op,0} \le \max_{0\le i<10} \sum_{j=0}^{9} \|B_{ij}\|_{op} \le \max_{0\le i<10} \sum_{j=0}^{9} \|B_{ij}\|_F < 31.13,$$

where the final inequality is a direct calculation using the exact rational entries of $(\widetilde{L} - I_{20})^{-1}$.

Each entry of $L_i - \widetilde{L}_i$ has absolute value less than 10^{-2}, so

$$\left\|L_i - \widetilde{L}_i\right\|_{op} \le \left\|L_i - \widetilde{L}_i\right\|_F < 2 \cdot 10^{-2}.$$

Because $L - \widetilde{L}$ has exactly one nonzero 2×2 block in each block row,

$$\left\|L - \widetilde{L}\right\|_{op,0} < 2 \cdot 10^{-2}.$$

Since $\left\|(\widetilde{L} - I_{20})^{-1}\right\|_{op,0}\left\|L - \widetilde{L}\right\|_{op,0} < 31.13 \cdot 2 \cdot 10^{-2} < 1$, the Neumann-series estimate gives

$$\kappa = \left\|(L - I_{20})^{-1}\right\|_{op,0} \le \frac{\left\|(\widetilde{L} - I_{20})^{-1}\right\|_{op,0}}{1 - \left\|(\widetilde{L} - I_{20})^{-1}\right\|_{op,0}\left\|L - \widetilde{L}\right\|_{op,0}} < \frac{31.13}{1 - 31.13 \cdot 2 \cdot 10^{-2}} < 83.$$

3.6. **Computing the mapping class of f^2 on a sphere with punctures.** In Lemma B.2, we demonstrate that radial projection $R : X(\mathbb{R}) \to S^2$, where $S^2 \subset \mathbb{R}^3$ denotes the unit sphere, is a diffeomorphism. In a small abuse of notation, we still use f to denote the associated conjugate diffeomorphism of S^2. Next, we stereographically project through $\gamma \coloneqq (0, 0, 1)$ to the x-y plane, which we view topologically as an open disk $\mathring{D}$. Let $\psi_\gamma : S^2 \setminus \{\gamma\} \to \mathbb{R}^2 \cong \mathring{D}$ denote stereographic projection through γ. Since $\gamma \notin \left\{f^k(x)\right\}_{k\in\mathbb{Z}}$, we can consider the image of $\left\{f^k(x)\right\}_{k\in\mathbb{Z}}$ under ψ_γ in $\mathring{D}$. The image is a set of ten points, which we label from left to right by $z_0, z_1, \ldots, z_9$, writing each as $z_i = (u_i, v_i)$ for $u_i < u_{i+1}$. Let τ be the permutation such that

$$z_{\tau(j)} = (\psi_\gamma \circ R)(f^j(x)).$$

Thus $\psi_\gamma \circ f \circ \psi_\gamma^{-1}$ maps $z_{\tau(j)}$ to $z_{\tau(j+1 \bmod 10)}$.

For each $0 \le i \le 8$, let $p_i : [0,1] \to \mathring{D}$ be the oriented straight-line arc from z_i to z_{i+1}, parametrized by $p_i(t) = (1-t)z_i + tz_{i+1}$. Let $\hat{z}_i$ and $\hat{p}_i$ denote the lifts of z_i and p_i, respectively, to S^2. Let $[\hat{s}_i] \in \mathrm{MOD}^+(S_{0,10})$ denote the clockwise half-twist around $\hat{p}_i$ (coming from the orientation induced on S^2 by the standard orientation in $\mathbb{R}^3$), and recall that the half-twists $[\hat{s}_i^{\pm}]$ generate $\mathrm{MOD}^+(S_{0,10})$. Our goal is to realize $[f^2] \in \mathrm{MOD}^+(S_{0,10})$ as a product of the $[\hat{s}_i^{\pm}]$'s.

Let $[s_i] \in \mathrm{MOD}^+(\mathring{D}_{10})$ denote the counter-clockwise half-twist about p_i, and notice that $[\psi_\gamma^{-1} \circ s_i \circ \psi_\gamma] = [\hat{s}_i]$ once we extend $\psi_\gamma^{-1} \circ s_i \circ \psi_\gamma$ to $S_{0,10}$ by fixing γ. Since f doesn't fix γ (although γ does have period 2), f doesn't directly descend to a homeomorphism of $\mathring{D}_{10}$. However, there exists $f' \in [f] \in \mathrm{MOD}^{\pm}(S_{0,10})$ such that $f'(\gamma) = \gamma$ and $f' \circ \hat{p}_i = f \circ \hat{p}_i$ as parametrized paths for all i. Indeed, this follows from the fact that the union $\cup_i \hat{p}_i$ does not separate S^2 and does not contain γ or $f^{-1}(\gamma)$. We use f' to denote the induced homeomorphism of $\mathring{D}_{10} \cong S_{0,11}$. Then

$$f' \circ p_i = \psi_\gamma \circ f \circ \hat{p}_i.$$

Let $S \coloneqq \{z_0, z_1, \dots, z_9\}$.

Definition 3.11. Let $T \subset \mathbb{R}^2$ be a finite set of marked points with distinct first coordinates, and let $\alpha : [0,1] \to \mathbb{R}^2$ be a continuous path whose endpoints lie in T and such that $\alpha(t) \notin T$ for all $t \in (0,1)$. Assume that there are only finitely many $t \in (0,1)$ for which $\alpha(t)$ lies on a vertical line through a point of T, and that at each such t, the path passes from one side of that vertical line to the other. The *arc-data of α with respect to T* is the sequence of marked points whose vertical lines α crosses, with repetitions recorded, together with whether α passes above or below each marked point.

Remark 3.12. If two paths with the same endpoints have the same arc-data with respect to T, then they are homotopic relative to T. Indeed, the vertical lines through the points of T divide $\mathbb{R}^2$ into simply connected regions. After sliding corresponding crossing points along the same components of these vertical lines, the portions of the two paths between corresponding crossings can be homotoped to one another within these regions.

Arc-data serves as an intermediate combinatorial tool for determining the relative homotopy classes of the paths $f'(p_i)$. Indeed, in Section C.4 we construct rational polygonal paths in a combinatorially isomorphic marked-point system and certify that their relative homotopy classes correspond to those of the paths $f'(p_i)$. We compute the arc-data of these polygonal paths exactly, perform homotopy-preserving simplifications, and use the resulting sequences as input to `mclass.c`. The top panel of Figure 4 depicts the certified relative homotopy classes of the paths $f'(p_i)$.

In what follows, we use $s_i.s_k$ to denote the composition $s_i \circ s_k$[1]. Using these simplified sequences, the algorithm described in Section 3.6.1 finds $i_k \in \{0,1,\dots,8\}$ and $n_k \in \{1,-1\}$ such that

$$g \coloneqq s_{i_0}^{n_0}.s_{i_1}^{n_1}.\dots.s_{i_\ell}^{n_\ell} \in \mathrm{Homeo}^+(\mathring{D}_{10})$$

maps each $f'(p_i)$ to p_i up to homotopy in $\mathring{D}_{10}$. This algorithm is implemented in K3entropy, `mclass.c`. The program is used to compute the words g_i displayed in Figure 4, but the proof does not rely on its correctness, since each word can be checked directly from the figure by tracing the indicated half-twists.

Then

$$\hat{g} \coloneqq \hat{s}_{i_0}^{n_0}.\hat{s}_{i_1}^{n_1}.\dots.\hat{s}_{i_\ell}^{n_\ell} \in \mathrm{Homeo}^+(S_{0,10} \setminus \{\gamma\})$$

maps each $f(\hat{p}_i)$ to $\hat{p}_i$ up to homotopy in $S_{0,11} = S_{0,10} \setminus \{\gamma\}$. Thus $[\hat{g} \circ f] \in \mathrm{MOD}^{\pm}(S_{0,10})$ is an orientation-reversing mapping class which fixes each $\hat{p}_i$; there is only one such mapping class. Indeed, let ν be the unique (up to homotopy in $S_{0,10}$) arc from $\hat{z}_9$ to $\hat{z}_0$ that's disjoint from each $\hat{p}_i$, and let L be the closed loop defined by the concatenation $\hat{p}_0.\hat{p}_1.\dots.\hat{p}_8.\nu$. Let $\mathcal{R} \in \mathrm{Homeo}(S_{0,10})$ be a reflection through L. Notice $\hat{g} \circ f \circ \mathcal{R}$ is orientation-preserving and preserves the homotopy class of each $\hat{p}_i$, so its mapping class is trivial. Thus $[\hat{g} \circ f] = [\mathcal{R}]$.

We are not quite done, since we wanted to realize $[f^2] \in \mathrm{MOD}^+(S_{0,10})$ as a product of the $[\hat{s}_i^{\pm}]$'s. Notice that $[\mathcal{R} \circ \hat{s}_i^{\pm}] = [\hat{s}_i^{\mp} \circ \mathcal{R}]$ for all i. Therefore

$$\begin{aligned} f^2 &\cong \hat{g}^{-1} \circ \mathcal{R} \circ \hat{g}^{-1} \circ \mathcal{R} \\ &\cong (\hat{s}_{i_\ell}^{-n_\ell}.\hat{s}_{i_{\ell-1}}^{-n_{\ell-1}}.\dots.\hat{s}_{i_0}^{-n_0}) \circ (\hat{s}_{i_\ell}^{n_\ell}.\hat{s}_{i_{\ell-1}}^{n_{\ell-1}}.\dots.\hat{s}_{i_0}^{n_0}) \\ &\cong \hat{g}^{-1} \circ \overline{g}. \end{aligned} \tag{3.13}$$

where $\overline{g}$ reverses the order of the product decomposition of $\hat{g}$, meaning

$$\overline{g} \coloneqq \hat{s}_{i_\ell}^{n_\ell}.\hat{s}_{i_{\ell-1}}^{n_{\ell-1}}.\dots.\hat{s}_{i_0}^{n_0}.$$

[1]This convention is consistent with the Flipper program.

3.6.1. *An algorithm for computing g.* Let f', z_i, p_i, and $s_i^{\pm}$ be as above, but consider the more general case of $\mathring{D}_n$. In what follows, we consider the z_i as marked points so that homeomorphisms of $\mathring{D}_n$ permute the marked points.

Let $\mathrm{MOD}_0^+(\mathring{D}_n) \subset \mathrm{MOD}^+(\mathring{D}_n)$ denote the subgroup generated by homotopy classes of homeomorphisms that fix the marked point z_0. The following lemma will be useful later:

Lemma 3.14. *The subgroup $MOD_0^+(\mathring{D}_n)$ satisfies*

$$MOD_0^+(\mathring{D}_n) = \langle [\tau_{p_0}], [s_1], [s_2], \dots, [s_{n-2}] \rangle$$

where $\tau_{p_0} = s_0.s_0$ is a full twist around p_0.

Proof of Lemma 3.14. For each $0 \le i < j \le n-1$, define

$$A_{i,j} \coloneqq (s_{j-1}.s_{j-2}.\dots.s_{i+1}).s_i^2.(s_{j-1}.s_{j-2}.\dots.s_{i+1})^{-1}.$$

Then $A_{i,j}$ represents a Dehn twist around some loop containing just z_i and z_j, and notice that $A_{0,1} = \tau_{p_0}$. Moreover, the collection $\{A_{i,j} \mid 0 \le i < j \le n-1\}$ generates the pure mapping class group $\mathrm{PMOD}^+(\mathring{D}_n)$ (see [12]). Therefore $\mathrm{PMOD}^+(\mathring{D}_n) \le \langle [\tau_{p_0}], [s_1], [s_2], \dots, [s_{n-2}] \rangle$. Finally, any $[\varphi] \in \mathrm{MOD}_0^+(\mathring{D}_n)$ can be written as a product of $[\varphi_1] \in \mathrm{PMOD}^+(\mathring{D}_n)$ and $[\varphi_2] \in \langle [s_1], [s_2], \dots, [s_{n-2}] \rangle$, proving the lemma. □

For all $0 \le i < n-1$, we will find $g_i \in \mathrm{Homeo}^+(\mathring{D}_n)$ such that $g_i(p_k) \cong p_k$ for each $0 \le k < i$ and $g_i \circ g_{i-1} \circ \dots \circ g_0 \circ f' \circ p_i \cong p_i$. Then $g \coloneqq g_{n-2} \circ g_{n-3} \circ \dots \circ g_0$ satisfies $g \circ f' \circ p_i \cong p_i$ for each $0 \le i < n-1$, as desired. We will determine g_i inductively in the following way.

Base case: Choose a finite sequence of half-twists $t_1, \dots, t_\ell \in \left\{ s_k^{\pm} \mid 0 \le k < n-1 \right\}$ such that

$$g_0 \coloneqq t_\ell.t_{\ell-1}.\dots.t_1 \in \mathrm{Homeo}^+(\mathring{D}_n)$$

satisfies $g_0 \circ f' \circ p_0 \cong p_0$. The half-twists t_k can be readily obtained from the arc-data of a representative of the relative homotopy class of $f' \circ p_0$. The process is straightforward and we omit details; one can follow along in Figures 3 and 4 to get a sense of it. The idea is to "drag" the endpoint of the representative backwards along the path towards its starting point using half-twists.

Inductive case: Let $i > 0$, and assume that g_k for $0 \le k < i$ have been constructed.

Step 1. Define $P_{i-1} \coloneqq p_0.p_1.\dots.p_{i-1}$. Define a homotopy $C : [0,1] \times \mathring{D} \to \mathring{D}$ such that:

- C_0 is the identity;
- $C_1(P_{i-1})$ is a point;
- C_1 restricts to a homeomorphism from $\mathring{D} \setminus P_{i-1}$ to $\mathring{D} \setminus \{C_1(P_{i-1})\}$;
- for all $t \in [0,1)$, C_t is a homeomorphism satisfying $C_t(P_{i-1}) \subset P_{i-1}$;
- for all $t \in [0,1]$, C_t fixes z_k whenever $k > i$.

Let $c \coloneqq C_1$. Considering $c(P_{i-1})$ to be a marked point along with the z_k's for $i < k \le n-1$, c defines a continuous map $c : \mathring{D}_n \to \mathring{D}_{n-i}$. We take generators $\{\mathfrak{s}_k \mid i \le k < n-1\}$ for $\mathrm{MOD}^+(\mathring{D}_{n-i})$, where $\mathfrak{s}_k$ is a counter-clockwise half-twist about $c(p_k)$. Notice that, for $i < k < n-1$, one has

$$\mathfrak{s}_k \circ c \cong c \circ s_k. \tag{3.15}$$

Moreover, the full twist $\tau_{c(p_i)} = \mathfrak{s}_i^2$ admits the following lift by c:

$$\tau_{c(p_i)} \circ c \cong c \circ (s_i.s_{i-1}.\dots.s_1.s_0.s_0.s_1.\dots.s_{i-1}.s_i). \tag{3.16}$$

Step 2. Consider the path $\alpha_i \coloneqq g_{i-1} \circ g_{i-2} \circ \dots \circ g_0 \circ f' \circ p_i : [0,1] \to \mathring{D}_n$ and $c(\alpha_i) \subset \mathring{D}_{n-i}$. Choose $\tilde{g}_i \in \mathrm{Homeo}^+(\mathring{D}_{n-i})$ such that $\tilde{g}_i(c(\alpha_i)) \cong c(p_i)$ in $\mathring{D}_{n-i}$. Notice that $c(\alpha_i)$ and $c(p_i)$ both have starting point $c(P_{i-1})$. Therefore, $[\tilde{g}_i] \in \mathrm{MOD}_0^+(\mathring{D}_{n-i})$, and by Lemma 3.14, $[\tilde{g}_i] \in \langle [\tau_{c(p_i)}], [\mathfrak{s}_{i+1}], [\mathfrak{s}_{i+2}], \dots, [\mathfrak{s}_{n-2}] \rangle$. Just like in the base case, a presentation for $[\tilde{g}_i]$ in terms of those generators can be obtained from the arc-data of a representative of the relative homotopy class of $c \circ \alpha_i$.

Step 3. By applying (3.15) and (3.16), we can lift $\tilde{g}_i$ to $g_i' \in \mathrm{Homeo}^+(\mathring{D}_n)$ satisfying $\tilde{g}_i \circ c \cong c \circ g_i'$. The homeomorphism g_i' fixes p_k whenever $0 \le k < i$. Moreover, g_i' maps α_i to p_i up to twisting around P_{i-1}; we make this precise in the next step.

Step 4.

Let $P_i \coloneqq P_{i-1}.p_i$, and let $\tau_{P_{i-1}}$ and τ_{P_i} denote the counter-clockwise Dehn twists about P_{i-1} and P_i, respectively. Choose representatives so that $P_{i-1}.g_i'(\alpha_i)$ is embedded. Then $g_i'(\alpha_i)$ and p_i have endpoints z_i and z_{i+1}, and

$$c(g_i'(\alpha_i)) \cong \tilde{g}_i(c(\alpha_i)) \cong c(p_i).$$

We use the following elementary fact.

Claim 3.17. *Suppose that β is an embedded arc from z_i to z_{i+1} satisfying $\beta \cap P_{i-1} = \{z_i\}$ and $c(\beta) \cong c(p_i)$ relative to the marked points of the collapsed disk. Then there exists $k \in \mathbb{Z}$ such that*

$$p_i \cong \tau_{P_{i-1}}^k(\beta).$$

Proof of Claim 3.17. Remove a small regular neighborhood of P_{i-1}. The arcs β and p_i become proper arcs beginning on the same boundary component. Since their images under c are homotopic, their only possible difference is an integer winding about this boundary component. This winding is generated by $\tau_{P_{i-1}}$, proving the claim. □

Applying Claim 3.17 with $\beta = g_i'(\alpha_i)$, we find that

$$p_i \cong \tau_{P_{i-1}}^k(g_i'(\alpha_i))$$

for some $k \in \mathbb{Z}$. One could let $g_i \coloneqq \tau_{P_{i-1}}^k \circ g_i'$, but we instead set $g_i \coloneqq \tau_{P_i}^{-k} \circ \tau_{P_{i-1}}^k \circ g_i'$. Notice that τ_{P_i} and $\tau_{P_{i-1}}$ commute; this can be seen by realizing τ_{P_i} and $\tau_{P_{i-1}}$ as twists around disjoint closed loops. So

$$\tau_{P_i}^{-k}\tau_{P_{i-1}}^k = (\tau_{P_i}^{-1}\tau_{P_{i-1}})^k = ((s_i^{-1}.\dots.s_0^{-1}).(s_0^{-1}.\dots.s_i^{-1}))^k,$$

and we have

$$g_i = ((s_i^{-1}.\dots.s_0^{-1}).(s_0^{-1}.\dots.s_i^{-1}))^k \circ g_i',$$

concluding the inductive step.

3.6.2. *Application to our case and use of Flipper.* In Figure 4, we apply the algorithm described in Section 3.6.1 to compute g. After applying $g_7 \circ \dots \circ g_0$, we have

$$g_7 \circ \dots \circ g_0 \circ f' \circ p_8 \cong p_8,$$

so we set $g_8 = \mathrm{id}$. We find:

$$\begin{aligned}
g &= g_8 \circ g_7 \circ g_6 \circ \dots \circ g_0 = g_7 \circ g_6 \circ \dots \circ g_0 \\
&= (s_8^{-1}.s_8^{-1}.s_7^{-1}.s_6^{-1}.s_5^{-1}.s_4^{-1}.s_3^{-1}.s_2^{-1}.s_1^{-1}.s_0^{-1}.s_0^{-1}.s_1^{-1}.s_2^{-1}.s_3^{-1}.s_4^{-1}.s_5^{-1}.s_6^{-1}.s_7^{-1}) \\
&\quad \circ (s_7.s_8^{-1}.s_8^{-1}.s_7^{-1}.s_6^{-1}.s_5^{-1}.s_4^{-1}.s_3^{-1}.s_2^{-1}.s_1^{-1}.s_0^{-1}.s_0^{-1}.s_1^{-1}.s_2^{-1}.s_3^{-1}.s_4^{-1}.s_5^{-1}.s_6^{-1}.s_7^{-1}) \\
&\quad \circ (s_6.s_7.s_8^{-1}.s_8^{-1}.s_7^{-1}.s_6^{-1}.s_5^{-1}.s_4^{-1}.s_3^{-1}.s_2^{-1}.s_1^{-1}.s_0^{-1}.s_0^{-1}.s_1^{-1}.s_2^{-1}.s_3^{-1}.s_4^{-1}.s_5^{-1}.s_6^{-1}.s_7) \\
&\quad \circ (s_5.s_6.s_7.s_8^{-1}.s_8^{-1}.s_7^{-1}.s_6^{-1}.s_5^{-1}.s_4^{-1}.s_3^{-1}.s_2^{-1}.s_1^{-1}.s_0^{-1}.s_0^{-1}.s_1^{-1}.s_2^{-1}.s_3^{-1}.s_4^{-1}.s_5^{-1}.s_6^{-1}.s_7^{-1}.s_8^{-1}) \\
&\quad \circ (s_4.s_5.s_6.s_7^{-1}.s_7^{-1}.s_6^{-1}.s_5^{-1}.s_4^{-1}.s_3^{-1}.s_2^{-1}.s_1^{-1}.s_0^{-1}.s_0^{-1}.s_1^{-1}.s_2^{-1}.s_3^{-1}.s_4^{-1}) \circ (s_3.s_4.s_5.s_6.s_7^{-1}.s_8^{-1}) \\
&\quad \circ (s_2.s_3.s_4.s_5.s_6^{-1}.s_7^{-1}.s_8^{-1}.s_8^{-1}.s_7) \circ (s_1.s_2.s_0.s_1.s_3.s_4.s_5^{-1}.s_6^{-1}.s_7^{-1}.s_8^{-1}.s_8^{-1}.s_7.s_6.s_5^{-1}).
\end{aligned}$$

As we saw in (3.13), $[f^2]$ is readily obtained from g. For completeness, we include its mapping class below:

$$\begin{aligned}
f^2 \cong\ & \big[(\hat{s}_5.\hat{s}_6^{-1}.\hat{s}_7^{-1}.\hat{s}_8.\hat{s}_8.\hat{s}_7.\hat{s}_6.\hat{s}_5.\hat{s}_4^{-1}.\hat{s}_3^{-1}.\hat{s}_1^{-1}.\hat{s}_0^{-1}.\hat{s}_2^{-1}.\hat{s}_1^{-1}) \circ (\hat{s}_7^{-1}.\hat{s}_8.\hat{s}_8.\hat{s}_7.\hat{s}_6.\hat{s}_5^{-1}.\hat{s}_4^{-1}.\hat{s}_3^{-1}.\hat{s}_2^{-1}) \\
&\quad \circ (\hat{s}_8.\hat{s}_7.\hat{s}_6^{-1}.\hat{s}_5^{-1}.\hat{s}_4^{-1}.\hat{s}_3^{-1}) \circ (\hat{s}_4.\hat{s}_3.\hat{s}_2.\hat{s}_1.\hat{s}_0.\hat{s}_0.\hat{s}_1.\hat{s}_2.\hat{s}_3.\hat{s}_4.\hat{s}_5.\hat{s}_6.\hat{s}_7.\hat{s}_7.\hat{s}_6^{-1}.\hat{s}_5^{-1}.\hat{s}_4^{-1}) \\
&\quad \circ (\hat{s}_8.\hat{s}_7.\hat{s}_6.\hat{s}_5.\hat{s}_4.\hat{s}_3.\hat{s}_2.\hat{s}_1.\hat{s}_0.\hat{s}_0.\hat{s}_1.\hat{s}_2.\hat{s}_3.\hat{s}_4.\hat{s}_5.\hat{s}_6.\hat{s}_7.\hat{s}_8.\hat{s}_8.\hat{s}_7^{-1}.\hat{s}_6^{-1}.\hat{s}_5^{-1}) \\
&\quad \circ (\hat{s}_7^{-1}.\hat{s}_6.\hat{s}_5.\hat{s}_4.\hat{s}_3.\hat{s}_2.\hat{s}_1.\hat{s}_0.\hat{s}_0.\hat{s}_1.\hat{s}_2.\hat{s}_3.\hat{s}_4.\hat{s}_5.\hat{s}_6.\hat{s}_7.\hat{s}_8.\hat{s}_8.\hat{s}_7^{-1}.\hat{s}_6^{-1}) \\
&\quad \circ (\hat{s}_7.\hat{s}_6.\hat{s}_5.\hat{s}_4.\hat{s}_3.\hat{s}_2.\hat{s}_1.\hat{s}_0.\hat{s}_0.\hat{s}_1.\hat{s}_2.\hat{s}_3.\hat{s}_4.\hat{s}_5.\hat{s}_6.\hat{s}_7.\hat{s}_8.\hat{s}_8.\hat{s}_7^{-1}) \\
&\quad \circ (\hat{s}_7.\hat{s}_6.\hat{s}_5.\hat{s}_4.\hat{s}_3.\hat{s}_2.\hat{s}_1.\hat{s}_0.\hat{s}_0.\hat{s}_1.\hat{s}_2.\hat{s}_3.\hat{s}_4.\hat{s}_5.\hat{s}_6.\hat{s}_7.\hat{s}_8.\hat{s}_8)\big] \\
&\circ \big[(\hat{s}_5^{-1}.\hat{s}_6.\hat{s}_7.\hat{s}_8^{-1}.\hat{s}_8^{-1}.\hat{s}_7^{-1}.\hat{s}_6^{-1}.\hat{s}_5^{-1}.\hat{s}_4.\hat{s}_3.\hat{s}_1.\hat{s}_0.\hat{s}_2.\hat{s}_1) \circ (\hat{s}_7.\hat{s}_8^{-1}.\hat{s}_8^{-1}.\hat{s}_7^{-1}.\hat{s}_6^{-1}.\hat{s}_5.\hat{s}_4.\hat{s}_3.\hat{s}_2) \\
&\quad \circ (\hat{s}_8^{-1}.\hat{s}_7^{-1}.\hat{s}_6.\hat{s}_5.\hat{s}_4.\hat{s}_3) \circ (\hat{s}_4^{-1}.\hat{s}_3^{-1}.\hat{s}_2^{-1}.\hat{s}_1^{-1}.\hat{s}_0^{-1}.\hat{s}_0^{-1}.\hat{s}_1^{-1}.\hat{s}_2^{-1}.\hat{s}_3^{-1}.\hat{s}_4^{-1}.\hat{s}_5^{-1}.\hat{s}_6^{-1}.\hat{s}_7^{-1}.\hat{s}_7^{-1}.\hat{s}_6.\hat{s}_5.\hat{s}_4) \\
&\quad \circ (\hat{s}_8^{-1}.\hat{s}_7^{-1}.\hat{s}_6^{-1}.\hat{s}_5^{-1}.\hat{s}_4^{-1}.\hat{s}_3^{-1}.\hat{s}_2^{-1}.\hat{s}_1^{-1}.\hat{s}_0^{-1}.\hat{s}_0^{-1}.\hat{s}_1^{-1}.\hat{s}_2^{-1}.\hat{s}_3^{-1}.\hat{s}_4^{-1}.\hat{s}_5^{-1}.\hat{s}_6^{-1}.\hat{s}_7^{-1}.\hat{s}_8^{-1}.\hat{s}_8^{-1}.\hat{s}_7.\hat{s}_6.\hat{s}_5) \\
&\quad \circ (\hat{s}_7.\hat{s}_6^{-1}.\hat{s}_5^{-1}.\hat{s}_4^{-1}.\hat{s}_3^{-1}.\hat{s}_2^{-1}.\hat{s}_1^{-1}.\hat{s}_0^{-1}.\hat{s}_0^{-1}.\hat{s}_1^{-1}.\hat{s}_2^{-1}.\hat{s}_3^{-1}.\hat{s}_4^{-1}.\hat{s}_5^{-1}.\hat{s}_6^{-1}.\hat{s}_7^{-1}.\hat{s}_8^{-1}.\hat{s}_8^{-1}.\hat{s}_7.\hat{s}_6) \\
&\quad \circ (\hat{s}_7^{-1}.\hat{s}_6^{-1}.\hat{s}_5^{-1}.\hat{s}_4^{-1}.\hat{s}_3^{-1}.\hat{s}_2^{-1}.\hat{s}_1^{-1}.\hat{s}_0^{-1}.\hat{s}_0^{-1}.\hat{s}_1^{-1}.\hat{s}_2^{-1}.\hat{s}_3^{-1}.\hat{s}_4^{-1}.\hat{s}_5^{-1}.\hat{s}_6^{-1}.\hat{s}_7^{-1}.\hat{s}_8^{-1}.\hat{s}_8^{-1}.\hat{s}_7) \\
&\quad \circ (\hat{s}_7^{-1}.\hat{s}_6^{-1}.\hat{s}_5^{-1}.\hat{s}_4^{-1}.\hat{s}_3^{-1}.\hat{s}_2^{-1}.\hat{s}_1^{-1}.\hat{s}_0^{-1}.\hat{s}_0^{-1}.\hat{s}_1^{-1}.\hat{s}_2^{-1}.\hat{s}_3^{-1}.\hat{s}_4^{-1}.\hat{s}_5^{-1}.\hat{s}_6^{-1}.\hat{s}_7^{-1}.\hat{s}_8^{-1}.\hat{s}_8^{-1})\big].
\end{aligned} \tag{3.18}$$

In K3entropy, `FlipperDilatation.ipynb`[2] proves the following Proposition. Recall the polynomial P defined in (1.4).

[2]Flipper uses right Dehn twists. The distinction between left and right is not important here, since left and right twists are conjugate by an orientation-reversing homeomorphism, and conjugation preserves topological entropy.

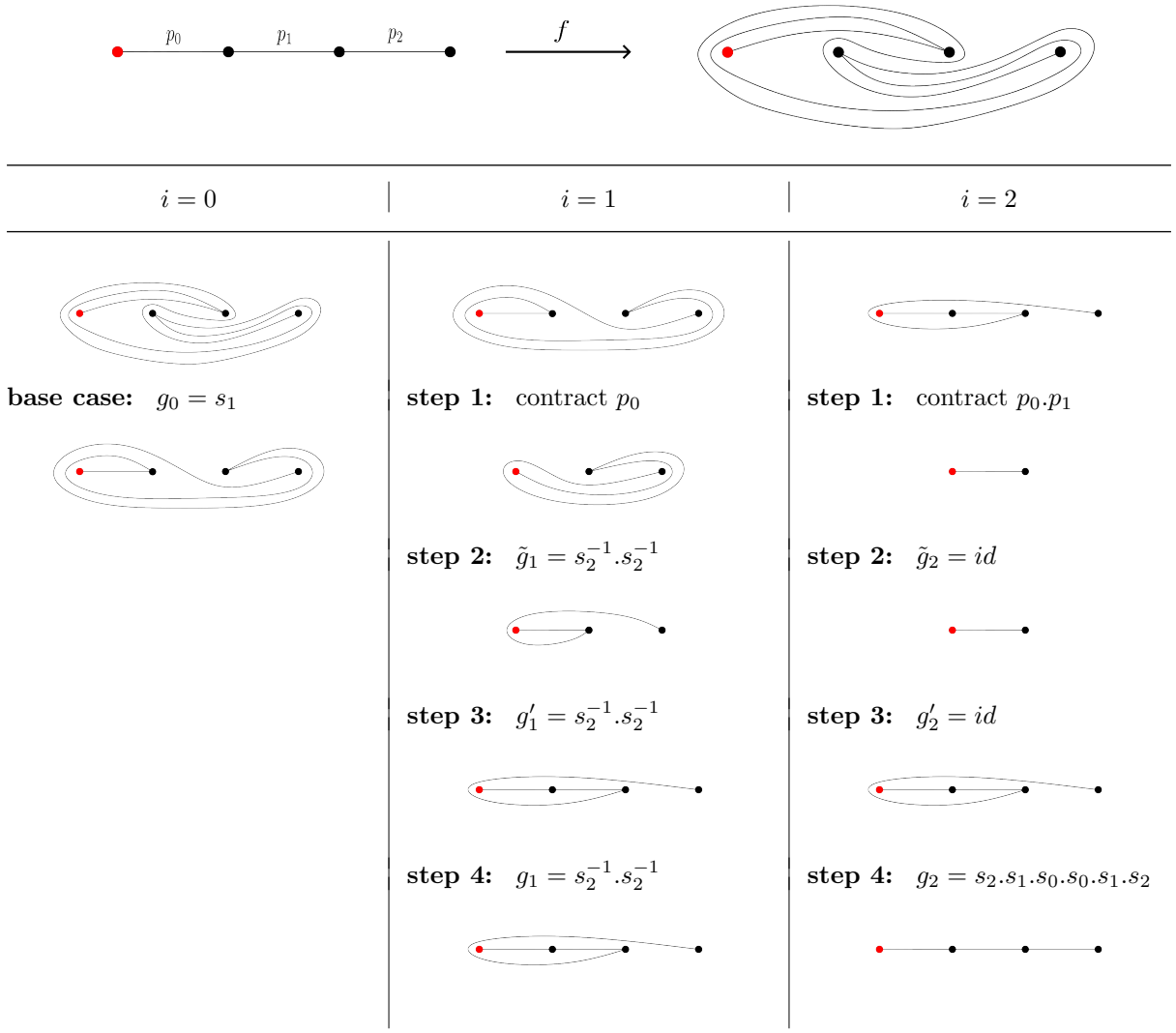


FIGURE 3. The algorithm described in Section 3.6.1 is applied in a simple case. We obtain $g = g_2 \circ g_1 \circ g_0 = s_2.s_1.s_0.s_0.s_1.s_2.s_2^{-1}.s_2^{-1}.s_1$.

Proposition 3.19 (Flipper certificate)**.** *The mapping class* $[f^2]$ *is pseudo-Anosov, and its stretching factor is the unique real root of* P *in* $(8.19981448, 8.19981449)$*, which we denote by* λ*.*

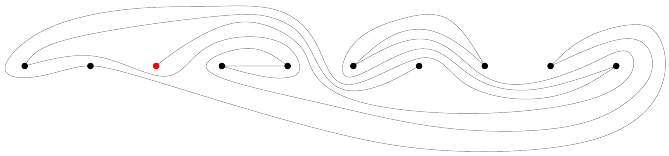
Proof. Flipper's Nielsen–Thurston classification routine verifies that the mapping class represented by the displayed word is pseudo-Anosov. It then constructs the 42×42 hitting matrix M for its invariant train track. The full matrix is printed by `FlipperDilatation.ipynb`, and its characteristic polynomial factors as

$$\chi_M(t) = P(t)(t-1)^4(t+1)^4(t^2+t+1)(t^{18}+t^{17}+\cdots+t+1).$$

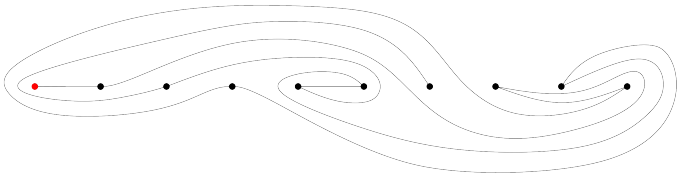
Then SageMath verifies that P is irreducible, and exact real-root isolation isolates all of its positive real roots. The largest is the unique root λ in $(8.19981448, 8.19981449)$, and P changes sign across the endpoints. Since M is nonnegative, its spectral radius is a positive real eigenvalue, and all the other factors in χ_M are cyclotomic. Therefore, λ is the spectral radius of M, and its minimal polynomial is P. This proves the claim. □

It follows that

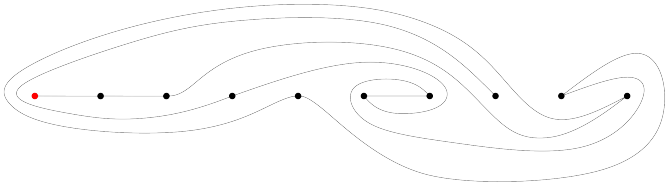
$$h_{top}(f, X(\mathbb{R})) \geq \frac{1}{2} \ln \lambda.$$



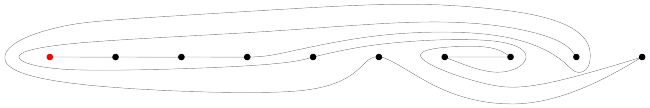
apply: $g_0 =$ $s_1.s_2.s_0.s_1.s_3.s_4.s_5^{-1}.s_6^{-1}.s_7^{-1}.s_8^{-1}.s_8^{-1}.s_7.s_6.s_5^{-1}$



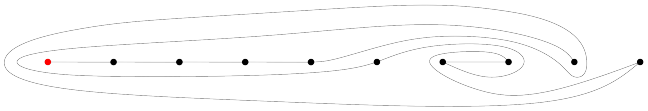
apply: $g_1 =$ $s_2.s_3.s_4.s_5.s_6^{-1}.s_7^{-1}.s_8^{-1}.s_8^{-1}.s_7$



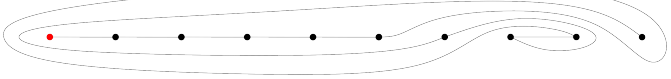
apply: $g_2 =$ $s_3.s_4.s_5.s_6.s_7^{-1}.s_8^{-1}$



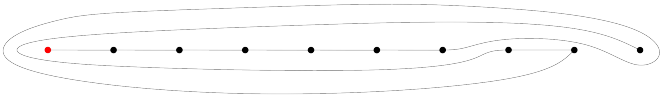
apply: $g_3 =$ $s_4.s_5.s_6.s_7^{-1}.s_7^{-1}.s_6^{-1}.s_5^{-1}.s_4^{-1}.s_3^{-1}.s_2^{-1}.s_1^{-1}.s_0^{-1}.s_0^{-1}.s_1^{-1}.s_2^{-1}.s_3^{-1}.s_4^{-1}$



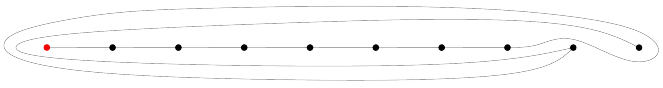
apply: $g_4 =$ $s_5.s_6.s_7.s_8^{-1}.s_8^{-1}.s_7^{-1}.s_6^{-1}.s_5^{-1}.s_4^{-1}.s_3^{-1}.s_2^{-1}.s_1^{-1}.s_0^{-1}.s_0^{-1}.s_1^{-1}.s_2^{-1}.s_3^{-1}.s_4^{-1}.s_5^{-1}.s_6^{-1}.s_7^{-1}.s_8^{-1}$



apply: $g_5 =$ $s_6.s_7.s_8^{-1}.s_8^{-1}.s_7^{-1}.s_6^{-1}.s_5^{-1}.s_4^{-1}.s_3^{-1}.s_2^{-1}.s_1^{-1}.s_0^{-1}.s_0^{-1}.s_1^{-1}.s_2^{-1}.s_3^{-1}.s_4^{-1}.s_5^{-1}.s_6^{-1}.s_7$



apply: $g_6 =$ $s_7.s_8^{-1}.s_8^{-1}.s_7^{-1}.s_6^{-1}.s_5^{-1}.s_4^{-1}.s_3^{-1}.s_2^{-1}.s_1^{-1}.s_0^{-1}.s_0^{-1}.s_1^{-1}.s_2^{-1}.s_3^{-1}.s_4^{-1}.s_5^{-1}.s_6^{-1}.s_7^{-1}$



apply: $g_7 =$ $s_8^{-1}.s_8^{-1}.s_7^{-1}.s_6^{-1}.s_5^{-1}.s_4^{-1}.s_3^{-1}.s_2^{-1}.s_1^{-1}.s_0^{-1}.s_0^{-1}.s_1^{-1}.s_2^{-1}.s_3^{-1}.s_4^{-1}.s_5^{-1}.s_6^{-1}.s_7^{-1}$

FIGURE 4. The top panel depicts representatives of the relative homotopy classes of the paths $f' \circ p_i$, as certified in Section C.4. The rest of the figure applies the algorithm described in Section 3.6.1 to the mapping class induced by f on the sphere punctured along the 10-periodic orbit obtained in Section 3.5.

Combining this with (2.5) gives

$$\frac{h_{top}(f, X(\mathbb{R}))}{h_{top}(f, X(\mathbb{C}))} \geq \frac{\ln \lambda}{2 \ln \mu} \approx 0.58 > \frac{1}{2}.$$

Appendices

A. Proof of the shadowing lemma

Proof of Lemma 3.2. To start, we set up a fixed point problem. Let $\varepsilon \coloneqq 12\kappa\delta$ so that $B_\varepsilon(0) \subset U_i$. Let

$$K \coloneqq \max_{0 \leq i < n} \left\| D^2 h_i^c \right\|_{C^0(B_\varepsilon(0)), \infty}.$$

Pick a smooth bump function $g : \mathbb{R}^2 \to \mathbb{R}$ satisfying $0 \leq g \leq 1$, $\operatorname{supp} g \subset B_\varepsilon(0)$, and $g\big|_{B_{\frac{\varepsilon}{2}}(0)} \equiv 1$. Furthermore, choose g such that $\|Dg\|_{C^0(\mathbb{R}^2)} < \frac{3}{\varepsilon}$. For each $0 \leq i < n$, let

$$L_i \coloneqq (Dh_i^c)_0.$$

On U_i, set

$$H_i \coloneqq g h_i^c + (1 - g) L_i.$$

Since $\operatorname{supp} g \subset B_\varepsilon(0) \subset U_i$, this extends to a C^2 map $H_i : \mathbb{R}^2 \to \mathbb{R}^2$ by setting $H_i = L_i$ outside U_i. Finally, let $M_i \coloneqq H_i - L_i : \mathbb{R}^2 \to \mathbb{R}^2$. Furthermore, define $H, L, M : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ by

$$\begin{aligned} H(w_0, \ldots, w_{n-1}) &\coloneqq (H_{n-1}(w_{n-1}), H_0(w_0), \ldots, H_{n-2}(w_{n-2})), \\ L(w_0, \ldots, w_{n-1}) &\coloneqq (L_{n-1}(w_{n-1}), L_0(w_0), \ldots, L_{n-2}(w_{n-2})), \\ M(w_0, \ldots, w_{n-1}) &\coloneqq (M_{n-1}(w_{n-1}), M_0(w_0), \ldots, M_{n-2}(w_{n-2})). \end{aligned}$$

We hope to find a fixed point $z = (z_0, \ldots, z_{n-1})$ for H in $B_{\frac{\varepsilon}{2}}(\vec{0})$. Here we are using the supremum norm $\|\cdot\|_0$ on $(\mathbb{R}^2)^n$ (see 1.4). Notice that we require $z_i \in B_{\frac{\varepsilon}{2}}(0)$ for every $0 \leq i < n$ in order to lift z to an h-periodic point. Indeed, since $H_i\big|_{B_{\frac{\varepsilon}{2}}(0)} \equiv h_i^c$, one has $H(z) = (h_{n-1}^c(z_{n-1}), h_0^c(z_0), \ldots, h_{n-2}^c(z_{n-2}))$. Thus $(\tilde{x}_i) \coloneqq (\varphi_i(z_i))$ is a fixed point for $(w_0, \ldots, w_{n-1}) \mapsto (h(w_{n-1}), h(w_0), \ldots, h(w_{n-2}))$ as a function $M^n \to M^n$. Note that

$$z = H(z) = L(z) + M(z)$$

exactly when

$$(L - id)z = -M(z)$$

or in other words

$$z = -(L - id)^{-1} M(z).$$

So, we hope to find a fixed point for $\mathcal{F} \coloneqq -(L - id)^{-1} M$ in $B_{\frac{\varepsilon}{2}}(\vec{0})$.

Claim A.1. *$\mathcal{F}$ is a $\frac{13}{16}$-contraction mapping for $\|\cdot\|_0$ whenever*

$$\varepsilon < \frac{1}{16\kappa K}.$$

We must prove $\sup_{w \in \mathbb{R}^{2n}} \left\| D((L - id)^{-1} M)_w \right\|_{op,0} \leq \frac{13}{16}$. We treat $\sup_{w \in \mathbb{R}^{2n}} \left\| D((L - id)^{-1})_w \right\|_{op,0}$ and $\sup_{w \in \mathbb{R}^{2n}} \|DM\|_{op,0}$ separately. Now L is linear, so $D((L - id)^{-1})_w = (L - id)^{-1}$ for all w. Thus $\sup_{w \in \mathbb{R}^{2n}} \left\| D((L - id)^{-1})_w \right\|_{op,0} = \kappa$.

Claim A.2. $\sup_{w \in \mathbb{R}^{2n}} \|(DM)_w\|_{op,0} \leq \frac{3\delta}{\varepsilon} + (6 + 2\sqrt{2})\varepsilon K < \frac{3\delta}{\varepsilon} + 9\varepsilon K$.

Assuming A.2, we have

$$\sup_{w \in \mathbb{R}^{2n}} \left\| D((L - id)^{-1} M)_w \right\|_{op,0} < \kappa \left(9 \cdot \frac{1}{16\kappa} + \frac{3}{12\kappa} \right) = \frac{13}{16},$$

proving A.1. To begin the proof of A.2, define $N_i^c : B_\varepsilon(0) \to \mathbb{R}^2$ by

$$N_i^c \coloneqq h_i^c - L_i.$$

The function N_i^c is essentially the non-linear part of h_i^c at 0, with the caveat that N_i^c might not fix the origin. On $B_\varepsilon(0)$, one has $M_i = H_i - L_i = g(h_i^c - L_i) = gN_i^c$, while M_i vanishes outside $\operatorname{supp} g$. Therefore,

$$\begin{aligned}\sup_{w\in\mathbb{R}^2}\|(DM_i)_w\|_{op} &\le \|DM_i\|_{C^0(\mathbb{R}^2)}\\ &\le \|Dg\|_{C^0(\mathbb{R}^2)}\|N_i^c\|_{C^0(B_\varepsilon(0))} + \|g\|_{C^0(\mathbb{R}^2)}\|DN_i^c\|_{C^0(B_\varepsilon(0))}\\ &\le \frac{3}{\varepsilon}\|N_i^c\|_{C^0(B_\varepsilon(0))} + \|DN_i^c\|_{C^0(B_\varepsilon(0))}.\end{aligned}$$

Thus

$$\begin{aligned}\sup_w\|(DM)_w\|_{op,0} &= \sup_w \max_i(\|(DM_i)_{w_i}\|_{op})\\ &\le \max_i\left(\frac{3}{\varepsilon}\|N_i^c\|_{C^0(B_\varepsilon(0))} + \|DN_i^c\|_{C^0(B_\varepsilon(0))}\right).\end{aligned}$$

So A.2 will follow from the following two subclaims.

Claim A.3. $\|N_i^c\|_{C^0(B_\varepsilon(0))} \le \delta + 2\varepsilon^2 K$.

Well, for all $v \in B_\varepsilon(0)$, Taylor's theorem implies that there exist $z_1, z_2 \in B_\varepsilon(0)$ such that

$$N_i^c(v) - h_i^c(0) = \frac{1}{2}\begin{pmatrix} v^t & 0\\ 0 & v^t\end{pmatrix}\cdot\begin{pmatrix}(D^2(h_i^c)_1)_{z_1} & 0\\ 0 & (D^2(h_i^c)_2)_{z_2}\end{pmatrix}\cdot\begin{pmatrix}v\\ v\end{pmatrix}$$

where $h_i^c = ((h_i^c)_1, (h_i^c)_2)$. Thus

$$\|N_i^c(v)\|_2 \le |v|^2 \max_{k=1,2}\left\|(D^2(h_i^c)_k)_{z_k}\right\|_F + \delta.$$

Since

$$\max_{k=1,2}\left\|(D^2(h_i^c)_k)_{z_k}\right\|_F \le 2\left\|D^2 h_i^c\right\|_{C^0(B_\varepsilon(0)),\infty}$$

and $|v| < \varepsilon$, we obtain

$$\|N_i^c\|_{C^0(B_\varepsilon(0))} \le \delta + 2\varepsilon^2 K.$$

Claim A.4. $\|DN_i^c\|_{C^0(B_\varepsilon(0)),op} \le 2\sqrt{2}\varepsilon K$.

Notice

$$(DN_i^c)_w = (Dh_i^c)_w - (Dh_i^c)_0.$$

By the mean value theorem applied to the four entries of Dh_i^c,

$$\|(DN_i^c)_w\|_{op} = \|(Dh_i^c)_w - (Dh_i^c)_0\|_{op} \le \|(Dh_i^c)_w - (Dh_i^c)_0\|_F \le 2\sqrt{2}\varepsilon K$$

Combining the two subclaims gives

$$\sup_{w\in\mathbb{R}^{2n}}\|(DM)_w\|_{op,0} \le \frac{3}{\varepsilon}\left(\delta + 2\varepsilon^2 K\right) + 2\sqrt{2}\varepsilon K = \frac{3\delta}{\varepsilon} + (6 + 2\sqrt{2})\varepsilon K < \frac{3\delta}{\varepsilon} + 9\varepsilon K,$$

proving Claim A.2, and hence Claim A.1. We are not quite done, since we need $z \in B_{\frac{\varepsilon}{2}}(\vec{0})$. Well, $z = \lim_{i\to\infty}\mathcal{F}^i(\hat{z})$ for any $\hat{z} \in \mathbb{R}^{2n}$. Therefore,

$$\begin{aligned}\|z - \hat{z}\|_0 &\le \sum_i \left\|\mathcal{F}^{i+1}(\hat{z}) - \mathcal{F}^i(\hat{z})\right\|_0\\ &\le \sum_i\left(\frac{13}{16}\right)^i\|\mathcal{F}(\hat{z}) - \hat{z}\|_0 = \frac{16}{3}\|\mathcal{F}(\hat{z}) - \hat{z}\|_0.\end{aligned}$$

In particular, substituting $\vec{0}$ for $\hat{z}$,

$$\begin{aligned}\|z\|_0 \le \frac{16}{3}\|\mathcal{F}(0)\|_0 &\le \frac{16}{3}\kappa\|M(0)\|_0\\ &= \frac{16}{3}\kappa\|H(0) - L(0)\|_0\\ &= \frac{16}{3}\kappa\|H(0)\|_0 < \frac{16}{3}\kappa\delta < 6\kappa\delta = \frac{\varepsilon}{2},\end{aligned}$$

concluding the proof. □

B. Miscellaneous

Lemma B.1. *For every nonzero $A \in \mathbb{R}$, the variety X_A is smooth.*

Proof. We first check that X_A is smooth on its affine part. Suppose that (x, y, z) were a singular point, meaning that $\partial_x q_A$, $\partial_y q_A$, and $\partial_z q_A$ vanish at (x, y, z). First, notice $xyz \neq 0$. Indeed, if, say, $x = 0$, then $\partial_x q_A = Ayz = 0$. After interchanging y and z if necessary, we may assume that $y = 0$, and then $\partial_z q_A = 2z = 0$. But $q_A(0, 0, 0)$ is nonzero, a contradiction. Moreover, none of $1 + x^2$, $1 + y^2$, and $1 + z^2$ vanishes. Indeed, if $1 + z^2 = 0$, then $\partial_x q_A = Ayz \neq 0$. Now

$$x\partial_x q_A - y\partial_y q_A = 2(x^2 - y^2)(1 + z^2),$$

and the analogous identities show that

$$x^2 = y^2 = z^2 = s.$$

Now $x\partial_x q_A = 0$ implies $Axyz = -2s(1+s)^2$. Substituting into $q_A = 0$, and squaring $Axyz = -2s(1+s)^2$, gives

$$s^3 + s^2 - s + 1 = 0, \qquad A^2 s = 4(1 + s)^4.$$

Eliminating s yields

$$A^6 - 48A^4 + 896A^2 + 1024 = 0,$$

which one can check has no real solutions.

Now assume that $x_0 = 0$, and set $x = 1$. On X_A, we then have

$$(y_0^2 + y^2)(z_0^2 + z^2) = 0.$$

If, say, $y_0^2 + y^2 = 0$ and $z_0^2 + z^2 \neq 0$, then $\partial_{y_0} \tilde{q}_A = 0$ implies $y_0 = 0$, and hence $y = 0$, a contradiction. If both factors vanish, then

$$\frac{\partial \tilde{q}_A}{\partial x_0} = Ay_0 y z_0 z \neq 0,$$

a contradiction. Hence X_A is smooth outside of the affine part. We conclude that X_A is smooth. □

Recall from Notation 3.1 that we let $X \coloneqq X_{10}$, $q \coloneqq q_{10}$, and $\tilde{q} \coloneqq \tilde{q}_{10}$.

Lemma B.2. *$X(\mathbb{R})$ is completely contained in the affine part of $(\mathbb{P}^1)^3$. Moreover, if $S^2 \subset \mathbb{R}^3$ denotes the unit sphere, then radial projection $R : X(\mathbb{R}) \to S^2$ is a diffeomorphism.*

Proof. In homogeneous coordinates, we find

$$\tilde{q}([0 : 1], [y_0 : y], [z_0 : z]) = (y_0^2 + y^2)(z_0^2 + z^2),$$

so, by the symmetry of $\tilde{q}$, $X(\mathbb{R})$ contains no points at infinity.

If every radial intersection, that is, an intersection of a ray from the origin with $X(\mathbb{R})$, were transverse, then $R : X(\mathbb{R}) \to S^2$ would be a local diffeomorphism. In that case, the image of R would be a nonempty subset of S^2 that is both closed and open, so R would be surjective. Moreover, since $X(\mathbb{R})$ is compact, R would be a covering map. Now the positive x-axis meets $X(\mathbb{R})$ exactly once, at $(1, 0, 0)$, since $q(t, 0, 0) = t^2 - 1$. Thus this covering would be one-sheeted and R would be a diffeomorphism.

For the sake of contradiction, assume there exists $p \in X(\mathbb{R})$ at which the radial intersection is tangent, or equivalently $p \in T_p X(\mathbb{R})$. In other words, $q(p) = 0$ and $Dq_p(p) = 0$, where

$$Dq_p(p) = x\frac{\partial q}{\partial x}(p) + y\frac{\partial q}{\partial y}(p) + z\frac{\partial q}{\partial z}(p).$$

Write $p = (x, y, z)$ and set $a = x^2$, $b = y^2$, $c = z^2$, $t = xyz$, and

$$s_1 = a + b + c, \qquad s_2 = ab + bc + ca.$$

Then $q(p) = 0$ gives

$$s_1 + s_2 + t^2 + 10t = 1, \tag{B.3}$$

and $Dq_p(p) = 0$ gives

$$2s_1 + 4s_2 + 6t^2 + 30t = 0. \tag{B.4}$$

Eliminating the term linear in t from (B.3) and (B.4) yields

$$s_1 = 3 + s_2 + 3t^2,$$

so

$$10t = -2(1 + s_2 + 2t^2). \tag{B.5}$$

Then (B.5) becomes

$$-5t = 1 + s_2 + 2t^2.$$

It remains to eliminate s_2. Now, since $ab, bc, ca \geq 0$, we can compare their arithmetic and geometric means to obtain

$$s_2 = ab + bc + ca \geq 3((ab)(bc)(ca))^{1/3} = 3t^{4/3}. \tag{B.6}$$

Hence

$$-5t \geq 1 + 3t^{4/3} + 2t^2.$$

Letting $u = t^{1/3}$, we have

$$1 + 5u^3 + 3u^4 + 2u^6 \leq 0.$$

But the above polynomial is positive for all u, a contradiction. □

Lemma B.7. *We have*

$$\max_{(x,y,z)\in X(\mathbb{R})} |x| < 2.31.$$

By symmetry, the same bound holds for the y and z coordinates.

Proof. Since $(x, y, z) \in X(\mathbb{R})$, we have $x, y \in \mathbb{R}$, and the discriminant of q as a function of z is nonnegative. Hence

$$100x^2y^2 - 4(1 + x^2)(1 + y^2)((1 + x^2)(1 + y^2) - 2) \geq 0.$$

Setting $s = 1 + y^2$ and $u = x^2$ yields

$$(1 + u)^2s^2 - (27u + 2)s + 25u \leq 0.$$

Now this is a quadratic polynomial in s, nonpositive at $s = 1 + y^2$, with positive leading coefficient. Hence it has a real zero and nonnegative discriminant. So

$$(27u + 2)^2 - 100u(1 + u)^2 \geq 0,$$

which expands to

$$100u^3 - 529u^2 - 8u - 4 \leq 0.$$

Now this polynomial in u has positive leading coefficient, and its largest zero is approximately 5.306. Hence $u < 5.31$, and $|x| < \sqrt{5.31} < 2.31$, concluding the lemma. □

Lemma B.8. *Let $S = \{(x, y) \in \mathbb{R}^2 \mid \mathcal{D}(x, y) > 0\}$. The function $\mathcal{D}$ satisfies the derivative bounds*

$$\max_{|\beta|=3} \sup_{(x,y)\in S} |D^\beta \mathcal{D}(x, y)| \leq 20000.$$

Proof. We find

$$\partial_{xxx}\mathcal{D}(x, y) = -96x(1 + y^2)^2$$

and

$$\partial_{xxy}\mathcal{D}(x, y) = 432y - 128x^2y(1 + y^2) - 64(1 + x^2)y(1 + y^2).$$

If $|x|, |y| \leq c$, we calculate

$$|\partial_{xxx}\mathcal{D}(x, y)| \leq 96c(1 + c^2)^2$$

and

$$|\partial_{xxy}\mathcal{D}(x, y)| \leq 432c + 128c^3(1 + c^2) + 64c(1 + c^2)^2.$$

The bound from Lemma B.7 and symmetry conclude the proof. □

Lemma B.9. *Considering f_A as a smooth function from an open neighborhood of $X_A(\mathbb{R})$ into $\mathbb{R}^3$, we have*

$$\left\|D^2 f_A\right\|_{C^0(X_A(\mathbb{R})),\infty} \leq 9^2(2 \cdot \max(|A|, 1)^2)^3$$

and

$$\|Df_A\|_{C^0(X_A(\mathbb{R})),\infty} \leq 3^2 \cdot \max(|A|, 1)^3.$$

Proof. Define

$$\alpha(x,y) \coloneqq \frac{xy}{(1+x^2)(1+y^2)}.$$

In affine coordinates, one calculates

$$\sigma_1^A(x,y,z) = (-x - A\alpha(y,z), y, z)\,, \qquad \sigma_2^A(x,y,z) = (x, -y - A\alpha(x,z), z)\,,$$

and

$$\sigma_3^A(x,y,z) = (x, y, -z - A\alpha(x,y))\,.$$

Furthermore, notice that

$$\alpha(x,y) = \frac{1}{4}\left(\frac{1}{-x-i} + \frac{1}{-x+i}\right)\left(\frac{1}{-y-i} + \frac{1}{-y+i}\right).$$

It follows that, for $x, y \in \mathbb{R}$,

$$|\alpha| \le \frac{1}{4}(1+1)(1+1) = 1, \qquad |\partial_y(\alpha)|, |\partial_x(\alpha)| \le \frac{1}{4}(1+1)(1+1) = 1$$

and

$$|\partial_{xy}(\alpha)| \le \frac{1}{4}(1+1)(1+1) = 1, \qquad |\partial_{yy}(\alpha)|, |\partial_{xx}(\alpha)| \le \frac{1}{4}2(1+1)(1+1) = 2.$$

Then $\left\|D\sigma_i^A\right\|_{X_A(\mathbb{R}),\infty} \le \max(1,|A|)$, and, by (B.11), $\left\|J^2(\sigma_i^A)\right\|_{X_A(\mathbb{R}),\infty} \le 2\cdot\max(|A|,1)^2$. The result follows from matrix multiplication and dimension counting. □

Lemma B.10. *If $C_1, C_2 \in \mathbb{R}_{>0}$ and $x, y \in \mathbb{R}$ are such that*

- $1 \le \mathcal{D}(x,y) \le C_1$ *and*
- $\left\|(D\mathcal{D})_{(x,y)}\right\|_\infty, \left\|(D^2\mathcal{D})_{(x,y)}\right\|_\infty \le C_2$

then

- $\left\|(D^2 p_\pm)_{(x,y)}\right\|_\infty \le 10 + C_1^{0.5} + \frac{3}{4}C_2 + \frac{1}{8}C_2^2$ *and*
- $\left\|(Dp_\pm)_{(x,y)}\right\|_\infty \le 5 + \frac{1}{2}C_1^{0.5} + \frac{1}{4}C_2$.

Proof. With α as in Lemma B.9 and $\beta(x,y) \coloneqq \frac{1}{(1+x^2)(1+y^2)}$,

$$p_\pm = -5\alpha \pm \frac{1}{2}\mathcal{D}^{\frac{1}{2}}\beta.$$

Notice that the same derivative bounds we established for α in Lemma B.9 hold for β. Then

$$\begin{aligned}|\partial_j p_\pm| &\le 5\,|\partial_j\alpha| + \frac{1}{2}\left(\left|\mathcal{D}^{0.5}\right|\,|\partial_j\beta| + |\beta|\left|\frac{\partial_j\mathcal{D}}{2\mathcal{D}^{0.5}}\right|\right)\\ &\le 5 + \frac{1}{2}C_1^{0.5} + \frac{1}{4}C_2\end{aligned}$$

and

$$\begin{aligned}|\partial_{ij} p_\pm| &\le 5\,|\partial_{ij}\alpha| + \frac{1}{2}\left(\left|\frac{\partial_i\mathcal{D}}{2\mathcal{D}^{0.5}}\right|\,|\partial_j\beta| + \left|\mathcal{D}^{0.5}\right|\,|\partial_{ij}\beta| + |\beta|\left|\frac{\partial_j\mathcal{D}\partial_i\mathcal{D}}{4\mathcal{D}^{1.5}}\right| + |\beta|\left|\frac{\partial_{ij}\mathcal{D}}{2\mathcal{D}^{0.5}}\right| + |\partial_i\beta|\left|\frac{\partial_j\mathcal{D}}{2\mathcal{D}^{0.5}}\right|\right)\\ &\le 10 + \frac{1}{2}\left(\frac{C_2}{2} + 2C_1^{0.5} + \frac{C_2^2}{4} + \frac{C_2}{2} + \frac{C_2}{2}\right)\\ &= 10 + C_1^{0.5} + \frac{3}{4}C_2 + \frac{1}{8}C_2^2,\end{aligned}$$

as desired. □

B.1. **Jets.** Consider $C^\infty(\mathbb{R}^n,\mathbb{R})$, the collection of smooth functions $f:\mathbb{R}^n\to\mathbb{R}$. Let $k\in\mathbb{N}$ and $x_0\in\mathbb{R}^n$, and consider the equivalence relation on $C^\infty(\mathbb{R}^n,\mathbb{R})$ defined by $f\sim g$ if $f(x_0)=g(x_0)$ and all partials of f and g at x_0 of order $\le k$ agree. We call such an equivalence class a k-jet, and denote the set of classes by $J^k_{x_0}(\mathbb{R}^n,\mathbb{R})$. Furthermore, denote by $J^k_{x_0}(\mathbb{R}^n,\mathbb{R})_0$ the collection of jets mapping x_0 to 0. By Taylor's theorem, $J^k_{x_0}(\mathbb{R}^n,\mathbb{R})_0$ is a linear space.

For a multi-index $\alpha = (\alpha_1,\dots,\alpha_n)\in\mathbb{Z}^n_{\ge 0}$, write

$$|\alpha| \coloneqq \sum_{j=1}^n \alpha_j, \qquad \alpha! \coloneqq \prod_{j=1}^n \alpha_j!, \qquad (x-x_0)^\alpha \coloneqq \prod_{j=1}^n \left(x_j - (x_0)_j\right)^{\alpha_j}.$$

The normalized monomials

$$\mathcal{B}_{x_0} \coloneqq \left\{ \frac{(x-x_0)^\alpha}{\alpha!} \;\middle|\; \alpha \in \mathbb{Z}^n_{\geq 0},\ 1 \leq |\alpha| \leq k \right\}$$

form a basis for $J^k_{x_0}(\mathbb{R}^n, \mathbb{R})_0$. We use $J^k(\mathbb{R}^n, \mathbb{R})_0$ to denote the trivial bundle obtained by assigning the vector space $J^k_{x_0}(\mathbb{R}^n, \mathbb{R})_0$ to each $x_0 \in \mathbb{R}^n$.

For any $f \in C^k(\mathbb{R}^n, \mathbb{R}^m)$ and $x_0 \in \mathbb{R}^n$, f induces a linear map $J^k_{x_0}(f) : J^k_{f(x_0)}(\mathbb{R}^m, \mathbb{R})_0 \longrightarrow J^k_{x_0}(\mathbb{R}^n, \mathbb{R})_0$ by precomposition. The entries of $J^k_{x_0}(f)$ depend on the partials of f at x_0 up to order k. Moreover, jets behave nicely under composition; precisely, whenever $f \circ g$ is defined, $J^k_{x_0}(f \circ g) = J^k_{x_0}(g) \cdot J^k_{g(x_0)}(f)$ in the sense of matrix multiplication. Often, we use $J^k(f)$ to denote the matrix-valued function $x_0 \mapsto J^k_{x_0}(f)$. For example, consider a smooth map $f : \mathbb{R}^2 \to \mathbb{R}^2$ with coordinate functions f_1, f_2. Order the basis $\mathcal{B}_{x_0}$ as

$$\left(x_1 - (x_0)_1,\ x_2 - (x_0)_2,\ \frac{(x_1-(x_0)_1)^2}{2},\ \frac{(x_2-(x_0)_2)^2}{2},\ (x_1-(x_0)_1)(x_2-(x_0)_2) \right),$$

and use the analogous ordering at $f(x_0)$. The 2-jet $J^2(f)$, expressed with respect to the bases $\mathcal{B}_{x_0}$ and $\mathcal{B}_{f(x_0)}$ on $J^2_{x_0}(\mathbb{R}^2, \mathbb{R})_0$ and $J^2_{f(x_0)}(\mathbb{R}^2, \mathbb{R})_0$, respectively, is

$$J^2(f) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_2}{\partial x_1} & 0 & 0 & 0 \\ \frac{\partial f_1}{\partial x_2} & \frac{\partial f_2}{\partial x_2} & 0 & 0 & 0 \\ \frac{\partial^2 f_1}{\partial x_1^2} & \frac{\partial^2 f_2}{\partial x_1^2} & \left(\frac{\partial f_1}{\partial x_1}\right)^2 & \left(\frac{\partial f_2}{\partial x_1}\right)^2 & 2\frac{\partial f_1}{\partial x_1}\frac{\partial f_2}{\partial x_1} \\ \frac{\partial^2 f_1}{\partial x_2^2} & \frac{\partial^2 f_2}{\partial x_2^2} & \left(\frac{\partial f_1}{\partial x_2}\right)^2 & \left(\frac{\partial f_2}{\partial x_2}\right)^2 & 2\frac{\partial f_1}{\partial x_2}\frac{\partial f_2}{\partial x_2} \\ \frac{\partial^2 f_1}{\partial x_1 \partial x_2} & \frac{\partial^2 f_2}{\partial x_1 \partial x_2} & \frac{\partial f_1}{\partial x_1}\frac{\partial f_1}{\partial x_2} & \frac{\partial f_2}{\partial x_1}\frac{\partial f_2}{\partial x_2} & \frac{\partial f_1}{\partial x_1}\frac{\partial f_2}{\partial x_2} + \frac{\partial f_2}{\partial x_1}\frac{\partial f_1}{\partial x_2} \end{pmatrix}.$$

In general, one has

$$\left\| J^2_{x_0}(f) \right\|_\infty \leq \max\left(\|(Df)_{x_0}\|_\infty, \left\|(D^2 f)_{x_0}\right\|_\infty, 2\|(Df)_{x_0}\|^2_\infty \right). \tag{B.11}$$

C. Computer-assisted certifications

C.1. **MPFR interval arithmetic.** The following interval-arithmetic framework, based on the GNU MPFR library [8], applies to `verifyPeriodicOrbit.c`, `derivativesInterval.c`, and `verifyArcs.c`. A nonzero finite MPFR number of precision N has the form

$$\pm 2^p \left(\frac{1}{2} + \sum_{k=2}^{N} c_k 2^{-k} \right),$$

where $c_k \in \{0, 1\}$ and p is an integer. The library allows users to specify an exponent range $[m, M]$. For fixed N and $[m, M]$, we call zero and the numbers of the displayed form with $p \in [m, M]$ *representable*. The smallest positive representable number is $2^m/2$, and the largest is $2^M(1 - 2^{-N})$. We say that a real number x is in the *allowable range* if $x = 0$ or

$$2^m/2 \leq |x| \leq 2^M(1 - 2^{-N}).$$

For a real number x in the allowable range, write $\mathrm{rd}(x)$ and $\mathrm{ru}(x)$ for the representable values obtained by rounding x toward $-\infty$ and $+\infty$, respectively. Each elementary MPFR operation takes allowable inputs and returns the exact real result rounded to a representable value according to the chosen mode, provided that the exact result lies in the allowable range. The modes `MPFR_RNDD` and `MPFR_RNDU` correspond to downward and upward rounding, respectively.

In this framework, a real number is represented by an interval containing it whose endpoints are representable. The programs then perform *outwardly rounded* interval arithmetic, which is standard interval arithmetic in which lower endpoints are rounded downward and upper endpoints are rounded upward. This ensures that every computed interval contains the corresponding exact result. Precisely, for intervals $I = [a, b]$ and $J = [c, d]$, the programs use the following:

$$I + J \subset [\mathrm{rd}(a + c), \mathrm{ru}(b + d)], \qquad I - J \subset [\mathrm{rd}(a - d), \mathrm{ru}(b - c)],$$

and

$$IJ \subset [\min_{s\in\{a,b\},\ t\in\{c,d\}} \mathrm{rd}(st), \max_{s\in\{a,b\},\ t\in\{c,d\}} \mathrm{ru}(st)].$$

If $0 \notin J$, it computes

$$J^{-1} \subset [\mathrm{rd}(1/d), \mathrm{ru}(1/c)], \qquad I/J = IJ^{-1}.$$

For $0 < a \leq b$, it computes

$$\sqrt{I} \subset [\mathrm{rd}(\sqrt{a}), \mathrm{ru}(\sqrt{b})].$$

Before performing interval division, the programs require the divisor interval to exclude zero. Before evaluating a square root, they require the interval under the radical to be nonnegative. If either requirement cannot be verified, the relevant program terminates.

C.2. **Certification in Sections 3.5.1 and 3.5.2** – `derivativesInterval.c`.

The program `derivativesInterval.c` uses the preceding interval operations with precision $N = 5000$. The program uses the polynomial formulas for $\mathcal{D}$ and its first and second partials to compute certified enclosures at each (a_i, b_i). These enclosures certify that the corresponding values displayed in Table 3 are correctly rounded to two decimal places. Then Table 3 gives us the bounds $K < 115$, $R < 163$ and $M < 442$.

The same evaluation on the square of radius 10^{-3} about each (a_i, b_i) certifies that $\mathcal{D} > 0$ there, and hence on $B_{10^{-3}}(a_i, b_i)$.

To estimate $L_i = (Df_i^c)_0$, the program uses explicit derivative formulas for the charts and the three involutions, combined with interval matrix multiplication, to certify enclosures for each entry of L_i. If $I_{i,jk}$ is the resulting enclosure of the (j,k)-entry of L_i, the program certifies

$$\max_{i,j,k} \sup_{x \in I_{i,jk}} |x - (\widetilde{L}_i)_{jk}| < 10^{-2}.$$

Hence $\left\| L_i - \widetilde{L}_i \right\|_\infty < 10^{-2}$, which is what we used in Section 3.5.2.

C.3. **Certifying the pseudo-orbit** – `verifyPeriodicOrbit.c`. The program `verifyPeriodicOrbit.c` uses the preceding interval operations with precision $N = 500$. The remaining certifications needed in Section 3.5 are that $0 \in U_i$ for every $0 \leq i \leq 9$ and that

$$\max_{0 \leq i \leq 9} \|f_i^c(0)\|_2 < 10^{-29}. \tag{C.1}$$

Recall that $\xi_i : \mathbb{R}^3 \to \mathbb{R}^2$ is the affine coordinate projection that inverts φ_i on $\varphi_i(V_i)$. For the remainder of this subsection, indices are taken modulo 10. Let $\tilde{\varphi}_i = \Psi_{j_i}^{\varepsilon_i}$ be the chart listed in Table 1. The program certifies a box containing $f\big(\tilde{\varphi}_i(a_i, b_i)\big)$ using the formulas for $\mathcal{D}$, $p_\pm$, and $\Psi_j^\pm$ from Section 3.4.1, and then applies $f = \sigma_3 \circ \sigma_2 \circ \sigma_1$.

Let $(u, v) \coloneqq \xi_{i+1}\big(f(\tilde{\varphi}_i(a_i, b_i))\big) + (a_{i+1}, b_{i+1})$ and let w be the coordinate lost in projection. The program first certifies that $\mathcal{D}(u, v) > 0$. It also checks that the image lies on the branch of $X(\mathbb{R})$ parametrized by $\tilde{\varphi}_{i+1}$. The branch is determined by the sign of

$$2(1 + u^2)(1 + v^2)w + 10uv.$$

The program certifies that its interval enclosure has the sign corresponding to $\tilde{\varphi}_{i+1}$. Thus $f(\varphi_i(0)) \in \varphi_{i+1}(V_{i+1})$, and hence $0 \in U_i$ for every i.

We now have

$$f_i^c(0) = \xi_{i+1}(f(\tilde{\varphi}_i(a_i, b_i))) \tag{C.2}$$

and intervals $R_{i,1}$ and $R_{i,2}$ satisfying $f_i^c(0) \in R_{i,1} \times R_{i,2}$.

For an interval $R = [\ell, u]$, define

$$m(R) \coloneqq \max\{|\ell|, |u|\}.$$

Using upward rounding, the program then computes

$$N_i \coloneqq \mathrm{ru}\left(\sqrt{m(R_{i,1})^2 + m(R_{i,2})^2}\right),$$

which bounds $\|f_i^c(0)\|_2$ from above.

Now Table 2 records the output of `verifyPeriodicOrbit.c`. Here $\mathcal{D}_i^{\mathrm{lb}}$ is a lower bound for the discriminant used to construct $\tilde{\varphi}_i(a_i, b_i)$, while $M_{i,k}$ and N_i satisfy

$$R_{i,k} \subset [-M_{i,k}, M_{i,k}] \quad \text{and} \quad \|f_i^c(0)\|_2 \leq N_i.$$

In Table 2, the displayed values of $\mathcal{D}_i^{\mathrm{lb}}$ have been rounded downward, while all other bounds have been rounded upward. In particular,

i	$\mathcal{D}_i^{\mathrm{lb}}$	$M_{i,1}$	$M_{i,2}$	N_i
0	114	$5.67 \cdot 10^{-32}$	$3.51 \cdot 10^{-31}$	$3.56 \cdot 10^{-31}$
1	8.70	$3.17 \cdot 10^{-31}$	$1.00 \cdot 10^{-30}$	$1.05 \cdot 10^{-30}$
2	70.2	$1.29 \cdot 10^{-30}$	$7.40 \cdot 10^{-31}$	$1.49 \cdot 10^{-30}$
3	8.59	$1.09 \cdot 10^{-30}$	$1.67 \cdot 10^{-31}$	$1.10 \cdot 10^{-30}$
4	108	$1.67 \cdot 10^{-31}$	$9.62 \cdot 10^{-32}$	$1.93 \cdot 10^{-31}$
5	10.4	$1.90 \cdot 10^{-31}$	$4.12 \cdot 10^{-31}$	$4.53 \cdot 10^{-31}$
6	108	$1.76 \cdot 10^{-31}$	$3.69 \cdot 10^{-31}$	$4.09 \cdot 10^{-31}$
7	8.59	$7.92 \cdot 10^{-31}$	$7.05 \cdot 10^{-30}$	$7.09 \cdot 10^{-30}$
8	70.2	$2.00 \cdot 10^{-30}$	$2.58 \cdot 10^{-30}$	$3.26 \cdot 10^{-30}$
9	8.70	$5.73 \cdot 10^{-31}$	$7.46 \cdot 10^{-31}$	$9.40 \cdot 10^{-31}$

TABLE 2. Certified interval bounds for the pseudo-orbit residuals.

$$\max_{0 \le i \le 9} \|f_i^c(0)\|_2 \le \max_{0 \le i \le 9} N_i \le 7.09 \cdot 10^{-30} < 10^{-29},$$

which proves (C.1).

C.4. **Certifying relative homotopy classes using arc-data – `verifyArcs.c`.** It remains, for $0 \le i \le 8$, to certify that $f' \circ p_i$ has the relative homotopy class represented by the corresponding arc in the top panel of Figure 4. In this section, we describe a strategy to do so, which is implemented in `verifyArcs.c`. The program `verifyArcs.c` uses the preceding interval operations with precision $N = 500$.

Recall from Section 3.6 that $\psi_\gamma : S^2 \setminus \{\gamma\} \to \mathbb{R}^2$ denotes stereographic projection through γ. Throughout this subsection, we work in the stereographic coordinate plane $\mathbb{R}^2$, identified topologically with $\mathring{D}$. Let $q_\gamma \coloneqq (\psi_\gamma \circ R \circ f^{-1} \circ R^{-1})(\gamma)$ and define $\Pi : \mathbb{R}^2 \setminus \{q_\gamma\} \to \mathbb{R}^2$ by

$$\Pi \coloneqq \psi_\gamma \circ R \circ f \circ R^{-1} \circ \psi_\gamma^{-1},$$

where $R : X(\mathbb{R}) \to S^2$ is the restriction to $X(\mathbb{R})$ of radial projection. Recall that f' is the homeomorphism of $\mathbb{R}^2$ obtained by stereographically conjugating a representative of $[f]$ on S^2 that fixes γ and whose restriction to each lifted arc $\hat{p}_i = \psi_\gamma^{-1} \circ p_i$ agrees with that of f. Since the arcs p_i avoid q_γ, it follows that $f' \circ p_i = \Pi \circ p_i$ for every $0 \le i \le 8$. Hence our program `verifyArcs.c` will work with the explicit map Π.

Recall from Section 3.6 that $z_0, \dots, z_9$ are the images under $\psi_\gamma \circ R$ of the points $f^j(x)$, indexed from left to right in the plane, and that τ is characterized by

$$z_{\tau(j)} = (\psi_\gamma \circ R)(f^j(x)).$$

The certification in `verifyArcs.c` happens in two steps:

Step 1. `verifyArcs.c` outputs the permutation τ and pairwise disjoint rational boxes $B_0, \dots, B_9 \subset \mathbb{R}^2$ with $z_i \in \mathrm{int}(B_i)$. The output boxes are certified to satisfy

$$\psi_\gamma \circ R \circ \varphi_j([-\bar{\eta}, \bar{\eta}]^2) \subset \mathrm{int}(B_{\tau(j)})$$

for all $0 \le j \le 9$, where $\bar{\eta} \coloneqq 5 \cdot 10^{-27}$ is as in Section 3.5. The program also certifies that every point of B_i lies strictly to the left of every point of B_{i+1} for $0 \le i \le 8$, and that $\gamma \notin R \circ \varphi_j([-\bar{\eta}, \bar{\eta}]^2)$ for every $0 \le j \le 9$. Consequently, the points z_i are pairwise distinct and x has least period ten.

Step 2. Fix $0 \le k \le 8$. Let $k_i, k_f \in \{0, \dots, 9\}$ be the indices determined by $\Pi(z_k) = z_{k_i}$ and $\Pi(z_{k+1}) = z_{k_f}$. Then `verifyArcs.c` finds an integer $m \ge 2$, rational numbers

$$0 = t_0 < t_1 < \cdots < t_m = 1,$$

and rectangles $R_0, \dots, R_{m-1}$. For $0 \le r < m$, set $I_r = [t_r, t_{r+1}]$. These satisfy the following properties:

(1) each R_r has rational vertices;
(2) for every $0 \le r < m$ and every $t \in I_r$, the map Π is defined on $(1-t)B_k + tB_{k+1}$ and
$$\Pi\big((1-t)B_k + tB_{k+1}\big) \subset R_r.$$
(3) for every $0 \le r < m$ and $0 \le \ell \le 9$,
$$R_r \cap B_\ell = \varnothing$$
unless $(r, \ell) = (0, k_i)$ or $(r, \ell) = (m-1, k_f)$;
(4) $B_{k_i} \subset R_0$ and $B_{k_f} \subset R_{m-1}$.

Each of the above properties is certified by the program.

By Lemma B.2, the radial inverse $R^{-1} : S^2 \to X(\mathbb{R})$ sends each $u \in S^2$ to the unique point $tu \in X(\mathbb{R})$ with $t > 0$. On each input box $U \subset \mathbb{R}^2$, to compute a box containing $\Pi(U)$, the program first certifies an interval $T \subset (0, \infty)$ containing, for every $v \in \psi_\gamma^{-1}(U)$, the unique positive solution t of $q(tv) = 0$. Multiplying a box containing $\psi_\gamma^{-1}(U)$ by T gives a box containing $R^{-1}(\psi_\gamma^{-1}(U))$. The program then applies f, R, and ψ_γ successively to this box using outward-rounded interval arithmetic to obtain a box containing $\Pi(U)$.

For each i, choose a rational reference point $\bar{z}_i \in \mathrm{int}(B_i) \cap \mathbb{Q}^2$. Let $\theta : \mathbb{R}^2 \to \mathbb{R}^2$ be a homeomorphism supported on $\cup_i B_i$ such that $\theta(z_i) = \bar{z}_i$.

Lemma C.3. *Assuming the existence of such I_r and R_r, we prove that $f' \circ p_k = \Pi \circ p_k$ is homotopic in $\mathbb{R}^2$ relative to $S = \{z_0, z_1, \ldots, z_9\}$ to $\theta^{-1} \circ L_k$ for some polygonal path L_k with vertices in $\mathbb{Q}^2$.*

Before proving Lemma C.3, we explain how it is used to certify the relative homotopy class of $f' \circ p_k$. Since $\theta : (\mathbb{R}^2, S) \to (\mathbb{R}^2, \theta(S))$ is a homeomorphism of pairs, θ^{-1} identifies the homotopy class of L_k relative to $\theta(S)$ with that of $\theta^{-1} \circ L_k$ relative to S. By Lemma C.3, the latter is the relative homotopy class of $f' \circ p_k$. Thus it suffices to determine the relative homotopy class of L_k. The program does this by computing its arc-data with respect to $\theta(S) = \{\bar{z}_0, \bar{z}_1, \ldots, \bar{z}_9\}$ exactly. This certified arc-data is used to illustrate the relative homotopy class of $f' \circ p_k$ in the top panel of Figure 4.

Proof of Lemma C.3. For each $0 \le r < m-1$, note that $R_r \cap R_{r+1}$ is a non-empty rational rectangle since it contains $\Pi\big((1-t_{r+1})B_k + t_{r+1}B_{k+1}\big)$. The program outputs a point $v_r \in R_r \cap R_{r+1} \cap \mathbb{Q}^2$ for every $0 \le r < m-1$. We let $v_{-1} \coloneqq \theta(\Pi(p_k(0)))$ and $v_{m-1} \coloneqq \theta(\Pi(p_k(1)))$. Define L_k on I_r, for $0 \le r < m$, to be the straight-line path from v_{r-1} to v_r. The points v_r are chosen so that L_k is embedded, as is also verified using exact rational arithmetic. The vertices v_r are also chosen so that L_k satisfies the hypotheses of Definition 3.11, as verified using exact rational arithmetic. Both v_{r-1} and v_r are contained in R_r, so by convexity, $L_k(I_r) \subset R_r$. Since θ is supported on $\bigcup_{\ell=0}^{9} B_\ell$, properties (3) and (4) give $\theta(R_r) \subset R_r$. Thus $(\theta \circ f' \circ p_k)(I_r) \subset R_r$ for $0 \le r < m$.

Using straight-line homotopies supported in the puncture-free intersections $R_r \cap R_{r+1}$, first homotope $\theta \circ f' \circ p_k$ to a path α such that $\alpha(t_{r+1}) = v_r$. For $1 \le r < m-1$, the rectangle R_r is puncture-free, so convexity of R_r gives a straight-line homotopy between $\alpha|_{I_r}$ and $L_k|_{I_r}$ relative to v_{r-1} and v_r. For $r = 0$ and $r = m-1$, the rectangle R_r contains no marked point other than the common marked endpoint of the two paths. Therefore, for $r \in \{0, m-1\}$, the paths $\alpha|_{I_r}$ and $L_k|_{I_r}$ are homotopic relative to v_{r-1} and v_r. Combining these homotopies gives a homotopy relative to $\theta(S)$ from $\theta \circ f' \circ p_k$ to L_k. Composing it with θ^{-1} gives a homotopy in $\mathbb{R}^2$ relative to S from $f' \circ p_k$ to $\theta^{-1} \circ L_k$, concluding the proof. □

Finally, `verifyArcs.c` computes the arc-data of each polygonal path exactly, performs homotopy-preserving simplifications of this arc-data, and verifies that both the original and simplified sequences agree with the corresponding rows stored in `arc_certificate_data.c`. The program `mclass.c` uses the simplified sequences.

D. Tables

i	$\mathcal{D}(a_i, b_i)$	$\partial_1\mathcal{D}(a_i, b_i)$	$\partial_2\mathcal{D}(a_i, b_i)$	$\partial_{11}\mathcal{D}(a_i, b_i)$	$\partial_{22}\mathcal{D}(a_i, b_i)$	$\partial_{12}\mathcal{D}(a_i, b_i)$
0	114.24	136.59	-45.98	-419.46	-441.48	-178.78
1	8.71	-22.00	15.52	39.45	13.84	-81.14
2	70.23	86.57	-124.38	-58.60	-2.96	-171.09
3	8.59	-40.85	4.26	113.57	-24.69	-87.82
4	108.31	39.04	162.94	-260.04	-171.83	27.57
5	10.43	23.28	-23.28	28.55	28.55	-92.99
6	108.31	-162.94	-39.04	-171.83	-260.04	27.57
7	8.59	-4.26	-40.85	-24.69	113.57	87.82
8	70.23	124.38	-86.57	-2.96	-58.60	-171.09
9	8.71	15.52	22.00	13.84	39.45	81.14

Table 3. Values used in the certified bounds for K, R, M

i	$\widetilde{L}_i$
0	$\begin{pmatrix} -0.2159 & -0.3755 \\ -0.4694 & 0.4623 \end{pmatrix}$
1	$\begin{pmatrix} 1.7718 & -2.1227 \\ -12.3247 & 16.3690 \end{pmatrix}$
2	$\begin{pmatrix} 0.3539 & -4.7730 \\ 0.4185 & -4.6570 \end{pmatrix}$
3	$\begin{pmatrix} -3.1432 & -0.4406 \\ -0.0916 & -1.1425 \end{pmatrix}$
4	$\begin{pmatrix} -0.4583 & 0.1150 \\ -0.6163 & 0.8316 \end{pmatrix}$
5	$\begin{pmatrix} 1.4772 & -1.9866 \\ 0.3707 & -2.6806 \end{pmatrix}$
6	$\begin{pmatrix} -0.8852 & 0.0258 \\ -0.1241 & 0.3218 \end{pmatrix}$
7	$\begin{pmatrix} 1.0118 & 1.1967 \\ 13.6472 & 13.3154 \end{pmatrix}$
8	$\begin{pmatrix} -0.6239 & -4.3400 \\ 0.7475 & 5.7641 \end{pmatrix}$
9	$\begin{pmatrix} -1.3601 & 1.6743 \\ -0.3730 & -2.2037 \end{pmatrix}$

TABLE 4. The matrices $\widetilde{L}_i$, chosen to satisfy $\widetilde{L}_i \approx (Df_i^c)_0$

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